# 2023 Annual Dairy Report Reissued WDR for Existing Milk Cow Dairies

## No: R5-2013-0122

Cottonwood Dairy

10561 W 140 HWY, Atwater, CA 95301

June 26, 2024

→ **The Power of Commitment**

**Annual Report - General Order No. R5-2007-0035**
*Reporting period 01/01/2023 to 12/31/2023.*

DAIRY FACILITY INFORMATION

**A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY:** Cottonwood Dairy

Physical address of dairy:

| Number and Street | City | County | Zip Code |
|---|---|---|---|
| 10561 W 140 HWY | Atwater | Merced | 95301 |

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 06/01/1980

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

XX56-XX16-XX12-XXXX XX56-XX16-XX13-XXXX XX56-XX17-XX11-XXXX XX56-XX18-XXX7-XXXX XX56-XX19-XX11-XXXX

**B. OPERATORS**

Company, Gallo Cattle

Operator name: Company, Gallo Cattle

Telephone no.: (209) 394-7984 (Landline) (Cellular)

| Mailing Address Number and Street | City | State | Zip Code |
|---|---|---|---|
| 10561 W Highway | Atwater | CA | 95301 |

**C. OWNERS**

Company, Gallo Cattle

Legal owner name: Company, Gallo Cattle

Telephone no.: (209) 394-7984 (Landline) (Cellular)

| Mailing Address Number and Street | City | State | Zip Code |
|---|---|---|---|
| 10561 W Highway | Atwater | CA | 95301 |

Gallo, Michael

Legal owner name: Gallo, Michael

Telephone no.: (209) 394-7984 (Landline) (Cellular)

| Mailing Address Number and Street | City | State | Zip Code |
|---|---|---|---|
| P.O. Box 775 | Atwater | CA | 95301 |

**This owner is responsible for paying permit fees.**

**Annual Report - General Order No. R5-2007-0035**
*Reporting period 01/01/2023 to 12/31/2023.*

AVAILABLE NUTRIENTS

## A. HERD INFORMATION

| | Milk Cows | Dry Cows | Bred Heifers (15-24 mo.) | Heifers (7-14 mo. to breeding) | Calves (4-6 mo.) | Calves (0-3 mo.) |
|---|---|---|---|---|---|---|
| Number open confinement | 1 | 0 | 0 | 0 | 0 | 0 |
| Number under roof | 0 | 0 | 0 | 0 | 0 | 0 |
| Maximum number | 1 | 0 | 0 | 0 | 0 | 0 |
| Average number | 0 | 0 | 0 | 0 | 0 | 0 |
| Avg live weight (lbs) | 2,000 | 0 | 0 | 0 | | |

Predominant milk cow breed: Jersey-Holstein Cross

Average milk production: 1 *pounds per cow per day*

## B. MANURE GENERATED

Total manure excreted by the herd: 1.00 *tons per reporting period*

Total nitrogen from manure: 1.00 *lbs per reporting period*

After ammonia losses (30% loss applied): 0.70 *lbs per reporting period*

Total phosphorus from manure: 1.00 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

## C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 205,067,268 *gallons*

Total nitrogen generated: 125,487.43 *lbs*

Total phosphorus generated: 14,116.86 *lbs*

Total potassium generated: 278,963.16 *lbs*

Total salt generated: 1,566,579.74 *lbs*

| | | |
|---|---|---|
| | 205,067,268 | *gallons applied* |
| + | 0 | *gallons exported* |
| - | 0 | *gallons imported* |
| = | 205,067,268 | *gallons generated* |

## D. FRESH WATER SOURCES

| Source Description | Type |
|---|---|
| MID | Surface water |
| Peck Drain | Surface water |
| Well 113.5 | Ground water |
| Well 130 | Ground water |
| Well 131 | Ground water |

| Source Description | Type |
|---|---|
| Well 132 | Ground water |
| Well 133 | Ground water |
| Well 134 | Ground water |
| Well 135 | Ground water |
| Well 136 | Ground water |
| Well 137 | Ground water |
| Well 139 | Ground water |
| Well 140 | Ground water |
| Well 144 | Ground water |
| Well 149 | Ground water |
| Well 80 | Ground water |
| Well 82 | Ground water |
| Well 91.5 | Ground water |
| Well 92 | Ground water |

## E. SUBSURFACE (TILE) DRAINAGE SOURCES

*No subsurface (tile) drainage sources entered.*

## F. NUTRIENT IMPORTS

*No dry manure nutrient imports entered.*

*No process wastewater nutrient imports entered.*

*No commercial or other nutrient imports entered.*

## G. NUTRIENT EXPORTS

*No solid nutrient exports entered.*

*No liquid nutrient exports entered.*

APPLICATION AREA

## A. LIST OF LAND APPLICATION AREAS

| Field name | Controlled acres | Cropable acres | Total harvests | Type of waste applied | Parcel number |
|---|---|---|---|---|---|
| CW-12 | 39 | 39 | 2 | process wastewater | XX56-XX19-XX12-XXXX |
| CW-18N | 42 | 42 | 1 | process wastewater | XX56-XX18-XXX8-XXXX |
| CW-18S | 38 | 38 | 1 | process wastewater | XX56-XX18-XXX8-XXXX |
| CW-19N | 70 | 70 | 1 | process wastewater | XX56-XX18-XXX8-XXXX |
| CW-19S | 55 | 55 | 2 | process wastewater | XX56-XX18-XXX8-XXXX<br>XX65-XXX1-XXX3-XXXX |
| CW-20 | 36 | 36 | 2 | process wastewater | XX65-XXX1-XXX3-XXXX |
| CW-21 | 61 | 61 | 2 | process wastewater | XX56-XX18-XXX8-XXXX<br>XX65-XXX1-XXX3-XXXX |
| CW-22 | 52 | 52 | 1 | process wastewater | XX56-XX18-XXX8-XXXX<br>XX65-XXX1-XXX3-XXXX |
| CW-23 | 36 | 36 | 1 | process wastewater | XX65-XXX1-XXX3-XXXX |
| CW-29 | 80 | 80 | 1 | process wastewater | XX65-XXX1-XXX3-XXXX |
| CW-30 | 41 | 41 | 2 | none | XX65-XXX1-XXX3-XXXX |
| CW-31 | 35 | 35 | 2 | none | XX65-XXX1-XXX3-XXXX |
| CW-9 | 20 | 20 | 2 | process wastewater | XX56-XX18-XXX7-XXXX |
| LP-1 | 65 | 65 | 2 | process wastewater | XX65-XXX1-XX13-XXXX |
| LP-2 | 45 | 45 | 1 | process wastewater | XX65-XXX1-XX13-XXXX |
| LP-3 | 23 | 23 | 1 | process wastewater | XX65-XXX1-XX13-XXXX |
| LP-4 | 10 | 10 | 1 | none | XX65-XXX1-XX12-XXXX<br>XX65-XXX1-XX13-XXXX |
| LP-5 | 8 | 8 | 1 | none | XX65-XXX1-XX13-XXXX |
| Totals for areas that were used for application | 756 | 756 | 26 | | |
| Totals for areas that were not used for application | | | | | |
| Land application area totals | 756 | 756 | 26 | | |

## B. CROPS AND HARVESTS

CW-12

Field name: CW-12

## CW-12

### 10/28/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough  Acres planted: 39  Plant date: 10/28/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/17/2023 | 704.00 *ton* | Dry-weight | | 60.4 | 11,500.00 | 3,200.00 | 23,400.00 | | 4.02 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 18.00 | 198.00 | 30.60 | 149.40 | 0.00 |
| Total actual harvest content | 18.05 | 164.41 | 45.75 | 334.54 | 574.72 |

### 06/20/2023: Corn, silage

Crop: Corn, silage  Acres planted: 39  Plant date: 06/20/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 10/14/2023 | 814.00 *ton* | Dry-weight | | 66.8 | 11,600.00 | 2,900.00 | 16,800.00 | | 4.11 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 20.87 | 160.76 | 40.19 | 232.83 | 569.60 |

## CW-18N

Field name: CW-18N

### 10/13/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough  Acres planted: 42  Plant date: 10/13/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/17/2023 | 738.00 *ton* | Dry-weight | | 50.1 | 9,500.00 | 2,100.00 | 18,800.00 | | 3.93 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 18.00 | 198.00 | 30.60 | 149.40 | 0.00 |
| Total actual harvest content | 17.57 | 166.59 | 36.83 | 329.68 | 689.18 |

## CW-18S

Field name: CW-18S

### 10/13/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough  Acres planted: 38  Plant date: 10/13/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/17/2023 | 768.00 *ton* | Dry-weight | | 50.1 | 9,500.00 | 2,100.00 | 18,800.00 | | 3.93 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 18.00 | 198.00 | 30.60 | 149.40 | 0.00 |
| Total actual harvest content | 20.21 | 191.62 | 42.36 | 379.20 | 792.69 |

## CW-19N

Field name: CW-19N

### 10/22/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough  Acres planted: 70  Plant date: 10/22/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/17/2023 | 1,293.00 *ton* | Dry-weight | | 58.6 | 9,600.00 | 2,600.00 | 25,900.00 | | 4.16 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 18.00 | 198.00 | 30.60 | 149.40 | 0.00 |
| Total actual harvest content | 18.47 | 146.83 | 39.77 | 396.12 | 636.24 |

## CW-19S

Field name: CW-19S

## CW-19S

### 10/21/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 55 Plant date: 10/21/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/17/2023 | 702.00 *ton* | Dry-weight | | 64.6 | 10,800.00 | 2,700.00 | 26,600.00 | | 4.28 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 18.00 | 198.00 | 30.60 | 149.40 | 0.00 |
| Total actual harvest content | 12.76 | 97.60 | 24.40 | 240.38 | 386.77 |

### 06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 55 Plant date: 06/14/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 10/14/2023 | 1,650.00 *ton* | Dry-weight | | 57.7 | 11,500.00 | 2,100.00 | 13,500.00 | | 3.22 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 30.00 | 291.87 | 53.30 | 342.63 | 817.24 |

## CW-20

Field name: CW-20

### 10/21/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 36 Plant date: 10/21/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/18/2023 | 681.00 *ton* | Dry-weight | | 58.0 | 10,300.00 | 2,100.00 | 14,500.00 | | 3.77 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 18.00 | 198.00 | 30.60 | 149.40 | 0.00 |
| Total actual harvest content | 18.92 | 163.67 | 33.37 | 230.41 | 599.05 |

## CW-20

### 06/14/2023: Corn, silage

Crop: Corn, silage  Acres planted: 36  Plant date: 06/14/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 10/14/2023 | 950.00 *ton* | Dry-weight | | 67.2 | 11,000.00 | 2,300.00 | 15,500.00 | | 3.21 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 26.39 | 190.42 | 39.82 | 268.32 | 555.69 |

## CW-21

Field name: CW-21

### 10/22/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough  Acres planted: 61  Plant date: 10/22/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/19/2023 | 1,066.00 *ton* | Dry-weight | | 72.3 | 13,100.00 | 3,300.00 | 28,400.00 | | 4.44 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 18.00 | 198.00 | 30.60 | 149.40 | 0.00 |
| Total actual harvest content | 17.48 | 126.83 | 31.95 | 274.95 | 429.85 |

### 06/16/2023: Corn, silage

Crop: Corn, silage  Acres planted: 61  Plant date: 06/16/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 10/18/2023 | 1,701.00 *ton* | Dry-weight | | 65.3 | 7,000.00 | 1,500.00 | 15,400.00 | | 3.70 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 27.89 | 135.47 | 29.03 | 298.03 | 716.04 |

## CW-22

Field name: CW-22

### 05/16/2023: Corn, silage

Crop: Corn, silage  Acres planted: 52  Plant date: 05/16/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/13/2023 | 1,100.00 *ton* | Dry-weight | | 65.3 | 11,000.00 | 1,500.00 | 15,400.00 | | 3.70 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 21.15 | 161.49 | 22.02 | 226.08 | 543.19 |

## CW-23

Field name: CW-23

### 05/19/2023: Corn, silage

Crop: Corn, silage  Acres planted: 36  Plant date: 05/19/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/13/2023 | 896.00 *ton* | Dry-weight | | 65.3 | 11,000.00 | 1,500.00 | 15,400.00 | | 3.70 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 24.89 | 190.00 | 25.91 | 266.00 | 639.10 |

## CW-29

Field name: CW-29

## CW-29

### 10/17/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough  Acres planted: 80  Plant date: 10/17/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/18/2023 | 1,315.00 *ton* | Dry-weight | | 39.2 | 7,200.00 | 2,200.00 | 15,900.00 | | 1.17 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 18.00 | 198.00 | 30.60 | 149.40 | 0.00 |
| Total actual harvest content | 16.44 | 143.91 | 43.97 | 317.81 | 233.86 |

## CW-30

Field name: CW-30

### 11/26/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough  Acres planted: 41  Plant date: 11/26/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/18/2023 | 343.00 *ton* | Dry-weight | | 63.0 | 10,400.00 | 3,100.00 | 21,800.00 | | 4.08 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 18.00 | 198.00 | 30.60 | 149.40 | 0.00 |
| Total actual harvest content | 8.37 | 64.38 | 19.19 | 134.96 | 252.58 |

### 06/19/2023: Corn, silage

Crop: Corn, silage  Acres planted: 41  Plant date: 06/19/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 10/16/2023 | 939.00 *ton* | Dry-weight | | 66.4 | 9,400.00 | 2,000.00 | 10,700.00 | | 3.46 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 22.90 | 144.67 | 30.78 | 164.68 | 532.51 |

## CW-31

Field name: CW-31

### 11/26/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough  Acres planted: 35  Plant date: 11/26/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/19/2023 | 263.00 *ton* | Dry-weight | | 60.1 | 13,100.00 | 3,100.00 | 25,000.00 | | 3.93 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 18.00 | 198.00 | 30.60 | 149.40 | 0.00 |
| Total actual harvest content | 7.51 | 78.55 | 18.59 | 149.91 | 235.66 |

### 06/20/2023: Corn, silage

Crop: Corn, silage  Acres planted: 35  Plant date: 06/20/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 10/16/2023 | 704.00 *ton* | Dry-weight | | 57.8 | 9,700.00 | 2,700.00 | 12,000.00 | | 3.43 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 20.11 | 164.67 | 45.84 | 203.72 | 582.29 |

## CW-9

Field name: CW-9

### 10/27/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough  Acres planted: 15  Plant date: 10/27/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/17/2023 | 302.00 *ton* | Dry-weight | | 51.1 | 9,400.00 | 2,500.00 | 19,700.00 | | 3.88 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 18.00 | 198.00 | 30.60 | 149.40 | 0.00 |
| Total actual harvest content | 20.13 | 185.09 | 49.23 | 387.90 | 763.99 |

## CW-9

### 06/14/2023: Corn, silage

Crop: Corn, silage  Acres planted: 15  Plant date: 06/14/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 10/14/2023 | 563.00 *ton* | Dry-weight | | 61.9 | 9,700.00 | 2,900.00 | 14,000.00 | | 4.21 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 37.53 | 277.42 | 82.94 | 400.41 | 1,204.08 |

## LP-1

Field name: LP-1

### 10/10/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough  Acres planted: 65  Plant date: 10/10/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/18/2023 | 991.00 *ton* | Dry-weight | | 59.2 | 13,100.00 | 3,300.00 | 29,200.00 | | 4.16 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 18.00 | 198.00 | 30.60 | 149.40 | 0.00 |
| Total actual harvest content | 15.25 | 162.98 | 41.05 | 363.27 | 517.54 |

### 07/09/2023: Corn, silage

Crop: Corn, silage  Acres planted: 65  Plant date: 07/09/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 11/06/2023 | 1,752.00 *ton* | Dry-weight | | 57.8 | 9,700.00 | 2,700.00 | 12,000.00 | | 3.43 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 26.95 | 220.67 | 61.42 | 272.99 | 780.29 |

## LP-2

Field name: LP-2

### 10/11/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 45 Plant date: 10/11/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/18/2023 | 540.00 *ton* | Dry-weight | | 59.2 | 7,000.00 | 2,100.00 | 15,800.00 | | 3.98 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 18.00 | 198.00 | 30.60 | 149.40 | 0.00 |
| Total actual harvest content | 12.00 | 68.54 | 20.56 | 154.71 | 389.72 |

## LP-3

Field name: LP-3

### 10/11/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 23 Plant date: 10/11/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/18/2023 | 229.00 *ton* | Dry-weight | | 60.7 | 9,900.00 | 3,000.00 | 20,400.00 | | 4.01 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 18.00 | 198.00 | 30.60 | 149.40 | 0.00 |
| Total actual harvest content | 9.96 | 77.48 | 23.48 | 159.65 | 313.82 |

## LP-4

Field name: LP-4

## LP-4

### 10/12/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough  Acres planted: 10  Plant date: 10/12/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/18/2023 | 100.00 *ton* | Dry-weight | | 60.7 | 9,900.00 | 3,000.00 | 20,400.00 | | 4.01 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 18.00 | 198.00 | 30.60 | 149.40 | 0.00 |
| Total actual harvest content | 10.00 | 77.81 | 23.58 | 160.34 | 315.19 |

## LP-5

Field name: LP-5

### 10/12/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough  Acres planted: 8  Plant date: 10/12/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/18/2023 | 100.00 *ton* | Dry-weight | | 60.7 | 9,900.00 | 3,000.00 | 20,400.00 | | 4.01 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 18.00 | 198.00 | 30.60 | 149.40 | 0.00 |
| Total actual harvest content | 12.50 | 97.27 | 29.48 | 200.43 | 393.98 |

NUTRIENT BUDGET

## A. LAND APPLICATIONS

CW-12 - 10/28/2022: Wheat, silage, soft dough

Field name: CW-12

Crop: Wheat, silage, soft dough

Plant date: 10/28/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 11/25/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 139 | Ground water | 1.01 | 0.00 | 0.00 | 314.95 | 6,751,800.00 *gal* |
| Application event totals | | 1.01 | 0.00 | 0.00 | 314.95 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 02/21/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 46.22 | 1,800,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 46.22 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 03/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 24.26 | 945,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 24.26 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/10/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| CW WSP #1 (M2) | Process wastewater | 49.34 | 8.06 | 178.96 | 1,205.81 | 1,624,000.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 46.91 | 1,827,000.00 *gal* |
| Application event totals | | 49.34 | 8.06 | 178.96 | 1,252.72 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/21/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 139 | Ground water | 1.01 | 0.00 | 0.00 | 314.95 | 6,751,800.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 78.80 | 3,069,000.00 *gal* |
| Application event totals | | 1.01 | 0.00 | 0.00 | 393.75 | |

## CW-12 - 06/20/2023: Corn, silage

Field name: CW-12

Crop: Corn, silage

Plant date: 06/20/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/05/2023 | Injection | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Pre-Fertilizer (Mid Valley Ag) | Liquid commercial fertilizer | 8.08 | 9.30 | 6.15 | 0.00 | |
| Application event totals | | 8.08 | 9.30 | 6.15 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 06/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 139 | Ground water | 0.74 | 0.00 | 0.00 | 231.82 | 4,969,800.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 58.00 | 2,259,000.00 *gal* |
| Application event totals | | 0.74 | 0.00 | 0.00 | 289.83 | |

| | | | | |
|---|---|---|---|---|
| 07/01/2023 | Sidedress | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Pre-Fertilizer (Nutrien) | Liquid commercial fertilizer | 44.00 | 0.00 | 0.00 | 0.00 | |
| Application event totals | | 44.00 | 0.00 | 0.00 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 07/11/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 82 | Ground water | 27.85 | 0.00 | 0.00 | 410.67 | 1,936,687.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 63.55 | 2,475,000.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 35.31 | 1,375,000.00 *gal* |
| Application event totals | | 27.85 | 0.00 | 0.00 | 509.53 | |

| | | | | |
|---|---|---|---|---|
| 07/25/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 82 | Ground water | 21.16 | 0.00 | 0.00 | 312.11 | 1,471,883.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 48.30 | 1,881,000.00 *gal* |
| Application event totals | | 21.16 | 0.00 | 0.00 | 360.41 | |

## CW-12 - 06/20/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/10/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 42.37 | 1,650,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 42.37 | |

| | | | | |
|---|---|---|---|---|
| 08/21/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 71.87 | 2,799,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 71.87 | |

| | | | | |
|---|---|---|---|---|
| 08/30/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 26.32 | 1,025,000.00 *gal* |
| MID | Surface water | 0.00 | 0.00 | 0.00 | 16.23 | 1,025,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 42.55 | |

| | | | | |
|---|---|---|---|---|
| 09/10/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 82 | Ground water | 30.58 | 0.00 | 0.00 | 450.99 | 2,126,835.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 69.79 | 2,718,000.00 *gal* |
| Application event totals | | 30.58 | 0.00 | 0.00 | 520.78 | |

| | | | | |
|---|---|---|---|---|
| 09/27/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 139 | Ground water | 1.04 | 0.00 | 0.00 | 323.26 | 6,930,000.00 *gal* |
| Application event totals | | 1.04 | 0.00 | 0.00 | 323.26 | |

## CW-18N - 10/13/2022: Wheat, silage, soft dough

Field name: CW-18N

Crop: Wheat, silage, soft dough

Plant date: 10/13/2022

## CW-18N - 10/13/2022: Wheat, silage, soft dough

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/18/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 72.84 | 3,055,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 72.84 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 12/12/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 104.43 | 4,380,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 104.43 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 01/29/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 45.51 | 5.33 | 94.50 | 616.79 | 4,873,260.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 103.72 | 4,350,000.00 *gal* |
| Application event totals | | 45.51 | 5.33 | 94.50 | 720.50 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/11/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 45.65 | 5.34 | 94.80 | 618.75 | 4,888,758.00 *gal* |
| Application event totals | | 45.65 | 5.34 | 94.80 | 618.75 | |

## CW-18S - 10/13/2022: Wheat, silage, soft dough

Field name: CW-18S

Crop: Wheat, silage, soft dough

Plant date: 10/13/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|

## CW-18S - 10/13/2022: Wheat, silage, soft dough

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 11/17/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 66.91 | 5.69 | 127.80 | 488.46 | 3,046,956.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 75.24 | 2,855,000.00 *gal* |
| Application event totals | | 66.91 | 5.69 | 127.80 | 563.70 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 12/15/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 86.70 | 3,290,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 86.70 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 01/25/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 49.12 | 5.75 | 102.00 | 665.75 | 4,759,116.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 165.23 | 6,270,000.00 *gal* |
| Application event totals | | 49.12 | 5.75 | 102.00 | 830.98 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/18/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 114.94 | 9.77 | 219.53 | 839.06 | 5,233,896.00 *gal* |
| Application event totals | | 114.94 | 9.77 | 219.53 | 839.06 | |

## CW-19N - 10/22/2022: Wheat, silage, soft dough

Field name: CW-19N

Crop: Wheat, silage, soft dough

Plant date: 10/22/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|

CW-19N - 10/22/2022: Wheat, silage, soft dough

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 11/28/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 79.55 | 6.76 | 151.94 | 580.71 | 6,672,750.00 *gal* |
| Application event totals | | 79.55 | 6.76 | 151.94 | 580.71 | |

| | | | | |
|---|---|---|---|---|
| 01/25/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 73.80 | 5,159,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 73.80 | |

| | | | | |
|---|---|---|---|---|
| 02/13/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 23.41 | 2.74 | 48.62 | 317.34 | 4,178,802.00 *gal* |
| Application event totals | | 23.41 | 2.74 | 48.62 | 317.34 | |

| | | | | |
|---|---|---|---|---|
| 03/17/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 37.12 | 3.16 | 70.90 | 270.99 | 3,113,868.00 *gal* |
| Application event totals | | 37.12 | 3.16 | 70.90 | 270.99 | |

| | | | | |
|---|---|---|---|---|
| 04/11/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 92 | Ground water | 17.08 | 0.00 | 0.00 | 537.85 | 3,444,000.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 30.79 | 2,152,500.00 *gal* |
| Application event totals | | 17.08 | 0.00 | 0.00 | 568.64 | |

| | | | | |
|---|---|---|---|---|
| 05/01/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 29.98 | 3.51 | 62.27 | 406.39 | 5,351,484.00 *gal* |
| Application event totals | | 29.98 | 3.51 | 62.27 | 406.39 | |

## CW-19S - 10/21/2022: Wheat, silage, soft dough

Field name: CW-19S

Crop: Wheat, silage, soft dough

Plant date: 10/21/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 11/05/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 139 | Ground water | 0.36 | 0.00 | 0.00 | 110.68 | 3,346,200.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 48.98 | 2,690,000.00 *gal* |
| Application event totals | | 0.36 | 0.00 | 0.00 | 159.66 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 02/05/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 47.70 | 5.65 | 61.41 | 159.87 | 1,228,032.00 *gal* |
| Application event totals | | 47.70 | 5.65 | 61.41 | 159.87 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 02/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 31.64 | 3.70 | 65.70 | 428.80 | 4,436,610.00 *gal* |
| Application event totals | | 31.64 | 3.70 | 65.70 | 428.80 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 03/27/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 39.40 | 7.13 | 81.82 | 534.01 | 5,525,160.00 *gal* |
| Application event totals | | 39.40 | 7.13 | 81.82 | 534.01 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/17/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| CW WSP #1 (M2) | Process wastewater | 10.86 | 0.92 | 20.75 | 79.30 | 716,000.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 48.80 | 2,680,000.00 *gal* |
| Application event totals | | 10.86 | 0.92 | 20.75 | 128.10 | |

## CW-19S - 06/14/2023: Corn, silage

## CW-19S - 06/14/2023: Corn, silage

Field name: CW-19S

Crop: Corn, silage

Plant date: 06/14/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/05/2023 | Injection | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Pre-Fertilizer (Mid Valley Ag) | Liquid commercial fertilizer | 8.08 | 9.30 | 6.15 | 0.00 | |
| Application event totals | | 8.08 | 9.30 | 6.15 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/16/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| CW WSP #1 (M1) | Process wastewater | 48.86 | 7.98 | 177.22 | 1,194.09 | 2,268,000.00 *gal* |
| Well 92 | Ground water | 15.86 | 0.00 | 0.00 | 499.29 | 2,512,000.00 *gal* |
| Application event totals | | 64.72 | 7.98 | 177.22 | 1,693.38 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/01/2023 | Sidedress | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Pre-Fertilizer (Nutrien) | Liquid commercial fertilizer | 44.00 | 0.00 | 0.00 | 0.00 | |
| Application event totals | | 44.00 | 0.00 | 0.00 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/05/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 92 | Ground water | 16.41 | 0.00 | 0.00 | 516.78 | 2,600,000.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 72.83 | 4,000,000.00 *gal* |
| Application event totals | | 16.41 | 0.00 | 0.00 | 589.61 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/23/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 92 | Ground water | 29.31 | 0.00 | 0.00 | 923.05 | 4,644,000.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 37.42 | 2,055,000.00 *gal* |
| Application event totals | | 29.31 | 0.00 | 0.00 | 960.47 | |

## CW-19S - 06/14/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/03/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 24.10 | 2.05 | 46.03 | 175.94 | 1,588,422.00 *gal* |
| Well 130 | Ground water | 10.37 | 0.00 | 0.00 | 208.11 | 1,258,360.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 35.14 | 1,930,000.00 *gal* |
| Application event totals | | 34.47 | 2.05 | 46.03 | 419.19 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 92 | Ground water | 16.46 | 0.00 | 0.00 | 518.37 | 2,608,000.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 73.01 | 4,010,000.00 *gal* |
| Application event totals | | 16.46 | 0.00 | 0.00 | 591.38 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/05/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 73.85 | 6.28 | 141.05 | 539.09 | 4,867,110.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 89.12 | 4,895,000.00 *gal* |
| Application event totals | | 73.85 | 6.28 | 141.05 | 628.21 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/18/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 66.32 | 5.64 | 126.67 | 484.13 | 4,370,928.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 30.04 | 1,650,000.00 *gal* |
| Application event totals | | 66.32 | 5.64 | 126.67 | 514.17 | |

## CW-20 - 10/21/2022: Wheat, silage, soft dough

Field name: CW-20

Crop: Wheat, silage, soft dough

Plant date: 10/21/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|

## CW-20 - 10/21/2022: Wheat, silage, soft dough

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 11/07/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 17.25 | 2.02 | 35.82 | 233.78 | 1,583,256.00 *gal* |
| Well 139 | Ground water | 0.64 | 0.00 | 0.00 | 200.11 | 3,960,000.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 55.63 | 2,000,000.00 *gal* |
| Application event totals | | 17.89 | 2.02 | 35.82 | 489.53 | |

| | | | | |
|---|---|---|---|---|
| 02/04/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 73.53 | 8.70 | 94.67 | 246.44 | 1,239,102.00 *gal* |
| Application event totals | | 73.53 | 8.70 | 94.67 | 246.44 | |

| | | | | |
|---|---|---|---|---|
| 03/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 65.86 | 5.60 | 125.79 | 480.76 | 2,841,054.00 *gal* |
| Application event totals | | 65.86 | 5.60 | 125.79 | 480.76 | |

| | | | | |
|---|---|---|---|---|
| 03/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 58.52 | 6.85 | 121.51 | 793.07 | 5,370,918.00 *gal* |
| Application event totals | | 58.52 | 6.85 | 121.51 | 793.07 | |

| | | | | |
|---|---|---|---|---|
| 04/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 92 | Ground water | 3.47 | 0.00 | 0.00 | 109.32 | 360,000.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 23.64 | 850,000.00 *gal* |
| Application event totals | | 3.47 | 0.00 | 0.00 | 132.96 | |

## CW-20 - 06/14/2023: Corn, silage

Field name: CW-20

Crop: Corn, silage

Plant date: 06/14/2023

CW-20 - 06/14/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/05/2023 | Injection | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Pre-Fertilizer (Mid Valley Ag) | Liquid commercial fertilizer | 8.08 | 9.30 | 6.15 | 0.00 | |
| Application event totals | | 8.08 | 9.30 | 6.15 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 06/17/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| CW WSP #1 (M1) | Process wastewater | 47.89 | 7.82 | 173.70 | 1,170.35 | 1,455,000.00 *gal* |
| Well 92 | Ground water | 22.45 | 0.00 | 0.00 | 706.93 | 2,328,000.00 *gal* |
| Application event totals | | 70.34 | 7.82 | 173.70 | 1,877.28 | |

| | | | | |
|---|---|---|---|---|
| 07/01/2023 | Sidedress | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Pre-Fertilizer (Nutrien) | Liquid commercial fertilizer | 44.00 | 0.00 | 0.00 | 0.00 | |
| Application event totals | | 44.00 | 0.00 | 0.00 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 07/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 92 | Ground water | 22.06 | 0.00 | 0.00 | 694.79 | 2,288,000.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 79.56 | 2,860,000.00 *gal* |
| Application event totals | | 22.06 | 0.00 | 0.00 | 774.34 | |

| | | | | |
|---|---|---|---|---|
| 07/26/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 92 | Ground water | 21.91 | 0.00 | 0.00 | 689.93 | 2,272,000.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 79.00 | 2,840,000.00 *gal* |
| Application event totals | | 21.91 | 0.00 | 0.00 | 768.93 | |

## CW-20 - 06/14/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 33.44 | 2.84 | 63.87 | 244.10 | 1,442,544.00 *gal* |
| Well 130 | Ground water | 24.95 | 0.00 | 0.00 | 500.81 | 1,982,080.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 84.56 | 3,040,000.00 *gal* |
| Application event totals | | 58.39 | 2.84 | 63.87 | 829.48 | |

| | | | | |
|---|---|---|---|---|
| 08/24/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 94.02 | 3,380,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 94.02 | |

| | | | | |
|---|---|---|---|---|
| 09/09/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 59.53 | 2,140,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 59.53 | |

| | | | | |
|---|---|---|---|---|
| 09/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 32.65 | 3.82 | 67.81 | 442.54 | 2,997,018.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 86.23 | 3,100,000.00 *gal* |
| Application event totals | | 32.65 | 3.82 | 67.81 | 528.77 | |

## CW-21 - 10/22/2022: Wheat, silage, soft dough

Field name: CW-21

Crop: Wheat, silage, soft dough

Plant date: 10/22/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|

## CW-21 - 10/22/2022: Wheat, silage, soft dough

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 11/21/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 42.31 | 3.60 | 80.81 | 308.86 | 3,092,712.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 59.92 | 3,650,000.00 *gal* |
| Application event totals | | 42.31 | 3.60 | 80.81 | 368.78 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 01/28/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 78.52 | 4,783,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 78.52 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/04/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 57.46 | 4.88 | 109.76 | 419.48 | 4,200,450.00 *gal* |
| Application event totals | | 57.46 | 4.88 | 109.76 | 419.48 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/26/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| CW WSP #1 (M2) | Process wastewater | 66.79 | 10.91 | 242.22 | 1,632.04 | 3,438,000.00 *gal* |
| Application event totals | | 66.79 | 10.91 | 242.22 | 1,632.04 | |

## CW-21 - 06/16/2023: Corn, silage

Field name: CW-21

Crop: Corn, silage

Plant date: 06/16/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/05/2023 | Injection | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Pre-Fertilizer (Mid Valley Ag) | Liquid commercial fertilizer | 8.08 | 9.30 | 6.15 | 0.00 | |
| Application event totals | | 8.08 | 9.30 | 6.15 | 0.00 | |

CW-21 - 06/16/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/19/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 67.25 | 5.72 | 128.44 | 490.90 | 4,915,572.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 101.29 | 6,170,000.00 *gal* |
| Application event totals | | 67.25 | 5.72 | 128.44 | 592.19 | |

| | | | | |
|---|---|---|---|---|
| 07/01/2023 | Sidedress | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Pre-Fertilizer (Nutrien) | Liquid commercial fertilizer | 44.00 | 0.00 | 0.00 | 0.00 | |
| Application event totals | | 44.00 | 0.00 | 0.00 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 07/11/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 98.42 | 5,995,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 98.42 | |

| | | | | |
|---|---|---|---|---|
| 07/28/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 86.93 | 5,295,500.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 86.93 | |

| | | | | |
|---|---|---|---|---|
| 08/12/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 83.48 | 5,085,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 83.48 | |

| | | | | |
|---|---|---|---|---|
| 08/25/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 99.65 | 6,070,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 99.65 | |

## CW-21 - 06/16/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 136 | Ground water | 3.73 | 0.00 | 0.00 | 247.64 | 4,200,000.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 98.50 | 6,000,000.00 *gal* |
| Application event totals | | 3.73 | 0.00 | 0.00 | 346.14 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 17.74 | 1.51 | 33.87 | 129.47 | 1,296,420.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 79.62 | 4,850,000.00 *gal* |
| Application event totals | | 17.74 | 1.51 | 33.87 | 209.09 | |

## CW-22 - 05/16/2023: Corn, silage

Field name: CW-22

Crop: Corn, silage

Plant date: 05/16/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/05/2023 | Injection | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Pre-Fertilizer (Mid Valley Ag) | Liquid commercial fertilizer | 8.08 | 9.30 | 6.15 | 0.00 | |
| Application event totals | | 8.08 | 9.30 | 6.15 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/16/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 38.06 | 4.45 | 79.03 | 515.80 | 5,045,706.00 *gal* |
| Application event totals | | 38.06 | 4.45 | 79.03 | 515.80 | |

CW-22 - 05/16/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/01/2023 | Sidedress | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Pre-Fertilizer (Nutrien) | Liquid commercial fertilizer | 44.00 | 0.00 | 0.00 | 0.00 | |
| Application event totals | | 44.00 | 0.00 | 0.00 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 06/12/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 41.81 | 4.89 | 86.81 | 649.32 | 5,542,626.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 122.48 | 6,360,000.00 *gal* |
| Application event totals | | 41.81 | 4.89 | 86.81 | 771.80 | |

| | | | | |
|---|---|---|---|---|
| 07/02/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| CW WSP #1 (M2) | Process wastewater | 34.05 | 5.56 | 123.48 | 831.96 | 1,494,000.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 31.97 | 1,660,000.00 *gal* |
| Application event totals | | 34.05 | 5.56 | 123.48 | 863.93 | |

| | | | | |
|---|---|---|---|---|
| 07/16/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 68.17 | 3,540,000.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 72.60 | 3,770,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 140.77 | |

| | | | | |
|---|---|---|---|---|
| 07/25/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 35.15 | 1,825,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 35.15 | |

## CW-22 - 05/16/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/06/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 85.41 | 4,435,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 85.41 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/17/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 65.86 | 3,420,000.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 32.93 | 1,710,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 98.79 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/28/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 31.11 | 2.64 | 59.42 | 227.09 | 1,938,480.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 67.59 | 3,510,000.00 *gal* |
| Application event totals | | 31.11 | 2.64 | 59.42 | 294.69 | |

## CW-23 - 05/19/2023: Corn, silage

Field name: CW-23

Crop: Corn, silage

Plant date: 05/19/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/19/2023 | Injection | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Pre-Fertilizer (Mid Valley Ag) | Liquid commercial fertilizer | 8.08 | 9.30 | 6.15 | 0.00 | |
| Application event totals | | 8.08 | 9.30 | 6.15 | 0.00 | |

## CW-23 - 05/19/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 43.21 | 5.06 | 89.73 | 585.66 | 3,966,258.00 *gal* |
| Application event totals | | 43.21 | 5.06 | 89.73 | 585.66 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/01/2023 | Sidedress | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Pre-Fertilizer (Nutrien) | Liquid commercial fertilizer | 44.00 | 0.00 | 0.00 | 0.00 | |
| Application event totals | | 44.00 | 0.00 | 0.00 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/17/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 45.06 | 1,620,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 45.06 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/06/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 36.86 | 1,325,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 36.86 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/18/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 65.40 | 7.65 | 135.81 | 886.39 | 6,002,892.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 67.87 | 2,440,000.00 *gal* |
| Application event totals | | 65.40 | 7.65 | 135.81 | 954.26 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/30/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 84.01 | 3,020,000.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 42.00 | 1,510,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 126.01 | |

## CW-23 - 05/19/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/10/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 41.03 | 1,475,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 41.03 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 136 | Ground water | 7.06 | 0.00 | 0.00 | 467.87 | 4,683,000.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 62.03 | 2,230,000.00 *gal* |
| Application event totals | | 7.06 | 0.00 | 0.00 | 529.90 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/30/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 59.66 | 5.07 | 113.95 | 435.51 | 2,573,652.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 84.28 | 3,030,000.00 *gal* |
| Application event totals | | 59.66 | 5.07 | 113.95 | 519.79 | |

## CW-29 - 10/17/2022: Wheat, silage, soft dough

Field name: CW-29

Crop: Wheat, silage, soft dough

Plant date: 10/17/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 11/08/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 43.58 | 4.77 | 84.72 | 744.79 | 8,321,688.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 83.37 | 6,660,000.00 *gal* |
| Application event totals | | 43.58 | 4.77 | 84.72 | 828.16 | |

## CW-29 - 10/17/2022: Wheat, silage, soft dough

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 02/01/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 14.68 | 1.72 | 30.47 | 198.90 | 2,993,328.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 55.20 | 4,410,000.00 *gal* |
| Application event totals | | 14.68 | 1.72 | 30.47 | 254.10 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 02/16/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 22.49 | 2.63 | 46.71 | 304.85 | 4,587,900.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 54.20 | 4,330,000.00 *gal* |
| Application event totals | | 22.49 | 2.63 | 46.71 | 359.05 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/05/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| CW WSP #1 (M2) | Process wastewater | 51.72 | 8.45 | 187.59 | 1,263.98 | 3,492,000.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 54.64 | 4,365,000.00 *gal* |
| Application event totals | | 51.72 | 8.45 | 187.59 | 1,318.62 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/24/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 35.03 | 4.10 | 72.74 | 474.77 | 7,145,070.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 115.29 | 9,210,000.00 *gal* |
| Application event totals | | 35.03 | 4.10 | 72.74 | 590.06 | |

## CW-30 - 11/26/2022: Wheat, silage, soft dough

Field name: CW-30

Crop: Wheat, silage, soft dough

Plant date: 11/26/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|

## CW-30 - 11/26/2022: Wheat, silage, soft dough

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 12/01/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 98.72 | 4,041,750.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 98.72 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/07/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 130 | Ground water | 11.28 | 0.00 | 0.00 | 226.38 | 1,020,380.00 *gal* |
| Well 136 | Ground water | 2.17 | 0.00 | 0.00 | 144.15 | 1,643,250.00 *gal* |
| Application event totals | | 13.45 | 0.00 | 0.00 | 370.53 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 61.45 | 2,516,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 61.45 | |

## CW-30 - 06/19/2023: Corn, silage

Field name: CW-30

Crop: Corn, silage

Plant date: 06/19/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/05/2023 | Injection | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Pre-Fertilizer (Mid Valley Ag) | Liquid commercial fertilizer | 8.08 | 9.30 | 6.15 | 0.00 | |
| Application event totals | | 8.08 | 9.30 | 6.15 | 0.00 | |

CW-30 - 06/19/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/21/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 140 | Ground water | 9.88 | 0.00 | 0.00 | 513.16 | 3,492,000.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 60.41 | 2,473,500.00 *gal* |
| Application event totals | | 9.88 | 0.00 | 0.00 | 573.58 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/01/2023 | Sidedress | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Pre-Fertilizer (Nutrien) | Liquid commercial fertilizer | 84.00 | 0.00 | 0.00 | 0.00 | |
| Application event totals | | 84.00 | 0.00 | 0.00 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/13/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 140 | Ground water | 11.64 | 0.00 | 0.00 | 604.86 | 4,116,000.00 *gal* |
| Application event totals | | 11.64 | 0.00 | 0.00 | 604.86 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/26/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 140 | Ground water | 15.45 | 0.00 | 0.00 | 802.37 | 5,460,000.00 *gal* |
| Application event totals | | 15.45 | 0.00 | 0.00 | 802.37 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/05/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 136 | Ground water | 6.42 | 0.00 | 0.00 | 425.55 | 4,851,000.00 *gal* |
| Application event totals | | 6.42 | 0.00 | 0.00 | 425.55 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 98.75 | 4,043,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 98.75 | |

## CW-30 - 06/19/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/30/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 98.88 | 4,048,500.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 98.88 | |

| | | | | |
|---|---|---|---|---|
| 09/15/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 108.92 | 4,459,500.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 108.92 | |

## CW-31 - 11/26/2022: Wheat, silage, soft dough

Field name: CW-31

Crop: Wheat, silage, soft dough Plant date: 11/26/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 130 | Ground water | 22.66 | 0.00 | 0.00 | 454.96 | 1,750,620.00 *gal* |
| Well 136 | Ground water | 4.37 | 0.00 | 0.00 | 289.71 | 2,819,250.00 *gal* |
| Application event totals | | 27.03 | 0.00 | 0.00 | 744.68 | |

| | | | | |
|---|---|---|---|---|
| 04/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 100.68 | 3,519,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 100.68 | |

## CW-31 - 06/20/2023: Corn, silage

Field name: CW-31

Crop: Corn, silage Plant date: 06/20/2023

## CW-31 - 06/20/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/05/2023 | Injection | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Pre-Fertilizer (Mid Valley Ag) | Liquid commercial fertilizer | 8.08 | 9.30 | 6.15 | 0.00 | |
| Application event totals | | 8.08 | 9.30 | 6.15 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 06/19/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 140 | Ground water | 13.68 | 0.00 | 0.00 | 710.62 | 4,128,000.00 *gal* |
| Application event totals | | 13.68 | 0.00 | 0.00 | 710.62 | |

| | | | | |
|---|---|---|---|---|
| 07/01/2023 | Sidedress | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Pre-Fertilizer (Nutrien) | Liquid commercial fertilizer | 84.00 | 0.00 | 0.00 | 0.00 | |
| Application event totals | | 84.00 | 0.00 | 0.00 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 07/10/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 140 | Ground water | 13.88 | 0.00 | 0.00 | 720.95 | 4,188,000.00 *gal* |
| Application event totals | | 13.88 | 0.00 | 0.00 | 720.95 | |

| | | | | |
|---|---|---|---|---|
| 07/26/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 136 | Ground water | 7.52 | 0.00 | 0.00 | 498.50 | 4,851,000.00 *gal* |
| Application event totals | | 7.52 | 0.00 | 0.00 | 498.50 | |

| | | | | |
|---|---|---|---|---|
| 08/05/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 133.83 | 4,677,500.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 133.83 | |

## CW-31 - 06/20/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/15/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 108.14 | 3,779,500.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 108.14 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/28/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 83.42 | 2,915,500.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 83.42 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/12/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 114.79 | 4,012,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 114.79 | |

## CW-9 - 10/27/2022: Wheat, silage, soft dough

Field name: CW-9

Crop: Wheat, silage, soft dough

Plant date: 10/27/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 12/21/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 86.52 | 1,296,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 86.52 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/03/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 110.82 | 1,660,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 110.82 | |

## CW-9 - 10/27/2022: Wheat, silage, soft dough

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/19/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 139 | Ground water | 1.40 | 0.00 | 0.00 | 435.16 | 3,588,000.00 *gal* |
| Application event totals | | 1.40 | 0.00 | 0.00 | 435.16 | |

## CW-9 - 06/14/2023: Corn, silage

Field name: CW-9

Crop: Corn, silage

Plant date: 06/14/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/05/2023 | Injection | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Pre-Fertilizer (Mid Valley Ag) | Liquid commercial fertilizer | 8.08 | 9.30 | 6.15 | 0.00 | |
| Application event totals | | 8.08 | 9.30 | 6.15 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/23/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| CW WSP #1 (M2) | Process wastewater | 145.75 | 23.81 | 528.61 | 3,561.73 | 1,845,000.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 123.17 | 1,845,000.00 *gal* |
| Application event totals | | 145.75 | 23.81 | 528.61 | 3,684.90 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/14/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 102.14 | 1,530,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 102.14 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/10/2023 | Shank | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Pre-Fertilizer (Nutrien) | Liquid commercial fertilizer | 44.00 | 0.00 | 0.00 | 0.00 | |
| Application event totals | | 44.00 | 0.00 | 0.00 | 0.00 | |

## CW-9 - 06/14/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/28/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 71.10 | 1,065,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 71.10 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/06/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 71.10 | 1,065,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 71.10 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/16/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 140.80 | 2,109,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 140.80 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/25/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 129.68 | 1,942,500.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 129.68 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/06/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 155.62 | 2,331,000.00 *gal* |
| MID | Surface water | 0.00 | 0.00 | 0.00 | 53.31 | 1,295,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 208.93 | |

## LP-1 - 10/10/2022: Wheat, silage, soft dough

Field name: LP-1

Crop: Wheat, silage, soft dough

Plant date: 10/10/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|

## LP-1 - 10/10/2022: Wheat, silage, soft dough

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/19/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 60.82 | 7.12 | 126.30 | 824.30 | 10,079,358.00 *gal* |
| Application event totals | | 60.82 | 7.12 | 126.30 | 824.30 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 12/16/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 57.31 | 6.71 | 119.01 | 776.72 | 9,497,568.00 *gal* |
| Application event totals | | 57.31 | 6.71 | 119.01 | 776.72 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 02/14/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| CW WSP #1 (M2) | Process wastewater | 27.62 | 4.51 | 100.17 | 674.92 | 1,515,000.00 *gal* |
| Application event totals | | 27.62 | 4.51 | 100.17 | 674.92 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 03/06/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 49.12 | 4.18 | 93.82 | 358.58 | 3,826,038.00 *gal* |
| Application event totals | | 49.12 | 4.18 | 93.82 | 358.58 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/19/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 136 | Ground water | 1.59 | 0.00 | 0.00 | 105.35 | 1,903,840.00 *gal* |
| Application event totals | | 1.59 | 0.00 | 0.00 | 105.35 | |

## LP-1 - 07/09/2023: Corn, silage

Field name: LP-1

Crop: Corn, silage

Plant date: 07/09/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|

LP-1 - 07/09/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/05/2023 | Injection | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Pre-Fertilizer (Mid Valley Ag) | Liquid commercial fertilizer | 8.08 | 9.30 | 6.15 | 0.00 | |
| Application event totals | | 8.08 | 9.30 | 6.15 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 07/18/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 130 | Ground water | 13.59 | 0.00 | 0.00 | 272.81 | 1,949,480.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 39.15 | 2,541,500.00 *gal* |
| Application event totals | | 13.59 | 0.00 | 0.00 | 311.96 | |

| | | | | |
|---|---|---|---|---|
| 07/25/2023 | Sidedress | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Pre-Fertilizer (Nutrien) | Liquid commercial fertilizer | 84.00 | 0.00 | 0.00 | 0.00 | |
| Application event totals | | 84.00 | 0.00 | 0.00 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 08/01/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 66.74 | 5.67 | 127.47 | 487.20 | 5,198,472.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 38.63 | 2,507,500.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 45.45 | 2,950,000.00 *gal* |
| Application event totals | | 66.74 | 5.67 | 127.47 | 571.28 | |

| | | | | |
|---|---|---|---|---|
| 08/18/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 130 | Ground water | 25.91 | 0.00 | 0.00 | 520.07 | 3,716,400.00 *gal* |
| Well 136 | Ground water | 4.99 | 0.00 | 0.00 | 331.17 | 5,985,000.00 *gal* |
| Application event totals | | 30.90 | 0.00 | 0.00 | 851.24 | |

## LP-1 - 07/09/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/02/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 74.95 | 4,865,000.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 63.71 | 4,135,250.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 138.66 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/18/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 90.13 | 5,850,000.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 76.61 | 4,972,500.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 166.73 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/09/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 60.00 | 7.02 | 124.59 | 813.17 | 9,943,320.00 *gal* |
| Application event totals | | 60.00 | 7.02 | 124.59 | 813.17 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/12/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 130 | Ground water | 9.55 | 0.00 | 0.00 | 191.60 | 1,369,200.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 32.35 | 2,100,000.00 *gal* |
| Application event totals | | 9.55 | 0.00 | 0.00 | 223.96 | |

## LP-2 - 10/11/2022: Wheat, silage, soft dough

Field name: LP-2

Crop: Wheat, silage, soft dough

Plant date: 10/11/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|

## LP-2 - 10/11/2022: Wheat, silage, soft dough

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 11/13/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 39.43 | 4.61 | 81.88 | 534.43 | 4,524,186.00 *gal* |
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 129.74 | 5,830,000.00 *gal* |
| Application event totals | | 39.43 | 4.61 | 81.88 | 664.17 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/30/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 80.39 | 3,612,500.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 80.39 | |

## LP-3 - 10/11/2022: Wheat, silage, soft dough

Field name: LP-3

Crop: Wheat, silage, soft dough Plant date: 10/11/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/27/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Cannery | Process wastewater | 49.79 | 5.83 | 103.39 | 674.76 | 2,919,528.00 *gal* |
| Application event totals | | 49.79 | 5.83 | 103.39 | 674.76 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/19/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Peck Drain | Surface water | 0.00 | 0.00 | 0.00 | 58.10 | 1,334,500.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 58.10 | |

## LP-4 - 10/12/2022: Wheat, silage, soft dough

Field name: LP-4

Crop: Wheat, silage, soft dough Plant date: 10/12/2022

LP-4 - 10/12/2022: Wheat, silage, soft dough

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 11/22/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 130 | Ground water | 17.73 | 0.00 | 0.00 | 355.84 | 391,200.00 *gal* |
| Application event totals | | 17.73 | 0.00 | 0.00 | 355.84 | |

LP-5 - 10/12/2022: Wheat, silage, soft dough

Field name: LP-5

Crop: Wheat, silage, soft dough

Plant date: 10/12/2022

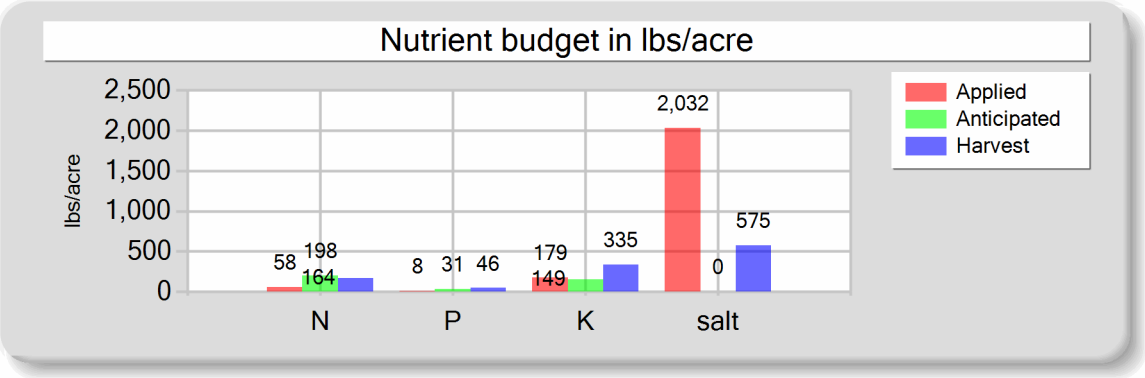
| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 11/22/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well 136 | Ground water | 4.34 | 0.00 | 0.00 | 287.96 | 640,500.00 *gal* |
| Application event totals | | 4.34 | 0.00 | 0.00 | 287.96 | |

## B. NUTRIENT BUDGET

CW-12 - 10/28/2022: Wheat, silage, soft dough

Field name: CW-12 Crop: Wheat, silage, soft dough Plant date: 10/28/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 49.34 | 8.06 | 178.96 | 1,205.81 |
| Fresh water | 2.02 | 0.00 | 0.00 | 826.09 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 58.37 | 8.06 | 178.96 | 2,031.90 |
| Anticipated crop nutrient removal | 198.00 | 30.60 | 149.40 | 0.00 |
| Actual crop nutrient removal | 164.41 | 45.75 | 334.54 | 574.72 |
| Nutrient balance | -106.04 | -37.69 | -155.58 | 1,457.17 |
| Applied to removed ratio | 0.36 | 0.18 | 0.53 | 3.54 |

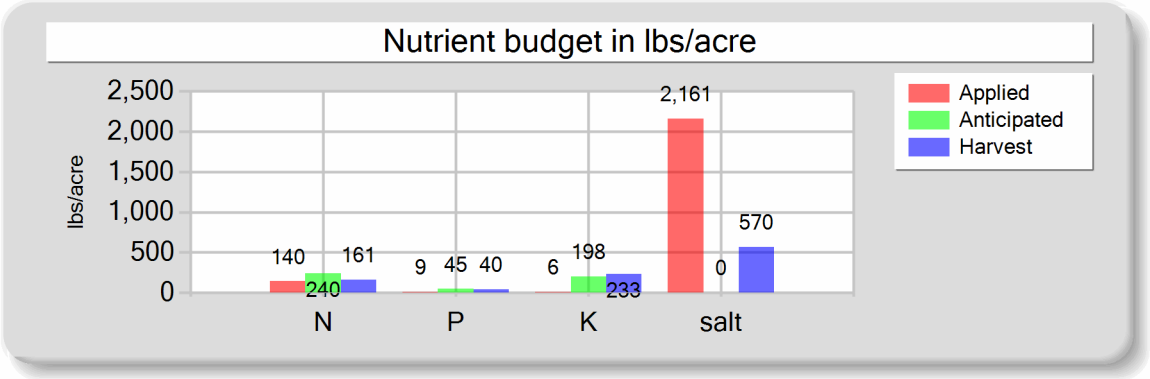
| Fresh water applied | |
|---|---|
| 21,144,600.00 | *gallons* |
| 778.68 | *acre-inches* |
| 19.97 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 1,624,000.00 | *gallons* |
| 59.81 | *acre-inches* |
| 1.53 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

CW-12 - 06/20/2023: Corn, silage

Field name: CW-12 Crop: Corn, silage Plant date: 06/20/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 52.08 | 9.30 | 6.15 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 81.38 | 0.00 | 0.00 | 2,160.59 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 140.46 | 9.30 | 6.15 | 2,160.59 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 160.76 | 40.19 | 232.83 | 569.60 |
| Nutrient balance | -20.31 | -30.89 | -226.68 | 1,590.99 |
| Applied to removed ratio | 0.87 | 0.23 | 0.03 | 3.79 |

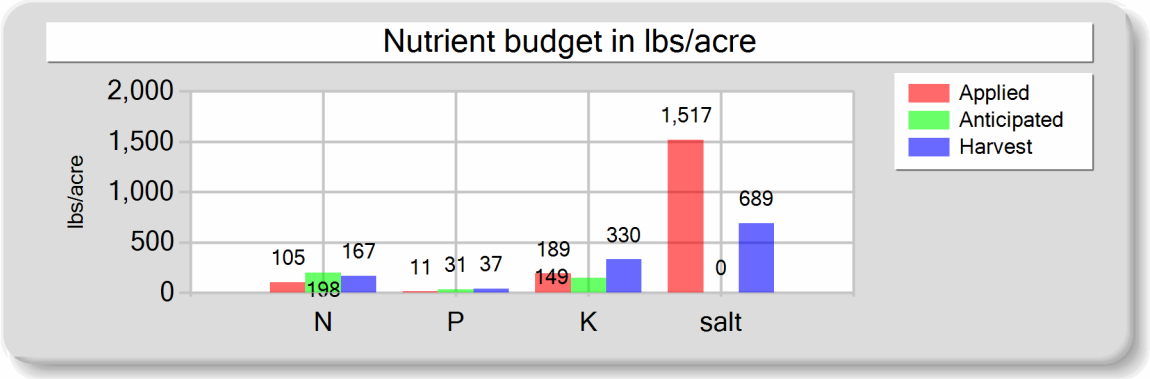
| Fresh water applied | |
|---|---|
| 34,642,205.00 | *gallons* |
| 1,275.75 | *acre-inches* |
| 32.71 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 0.00 | *gallons* |
| 0.00 | *acre-inches* |
| 0.00 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

CW-18N - 10/13/2022: Wheat, silage, soft dough

Field name: CW-18N  Crop: Wheat, silage, soft dough  Plant date: 10/13/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 91.16 | 10.67 | 189.31 | 1,235.54 |
| Fresh water | 0.00 | 0.00 | 0.00 | 280.99 |
| Atmospheric deposition | 14.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 105.16 | 10.67 | 189.31 | 1,516.53 |
| Anticipated crop nutrient removal | 198.00 | 30.60 | 149.40 | 0.00 |
| Actual crop nutrient removal | 166.59 | 36.83 | 329.68 | 689.18 |
| Nutrient balance | -61.43 | -26.16 | -140.38 | 827.35 |
| Applied to removed ratio | 0.63 | 0.29 | 0.57 | 2.20 |

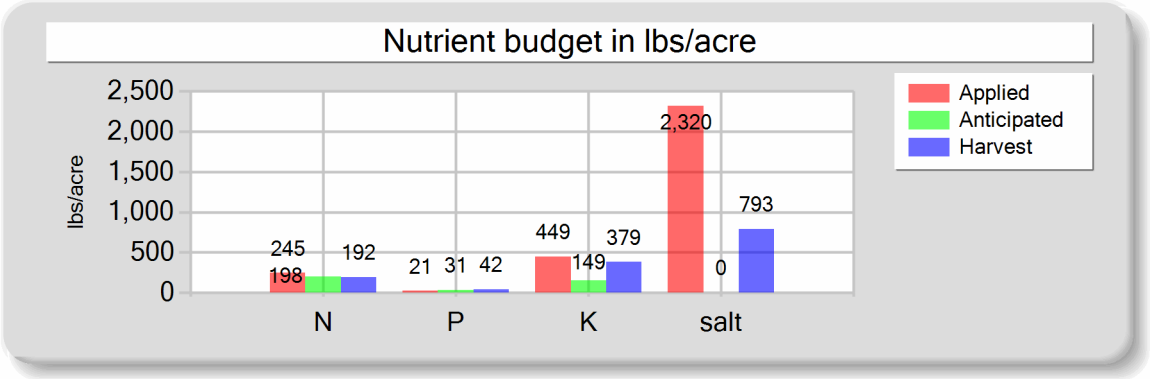
| Fresh water applied | |
|---|---|
| 11,785,000.00 | *gallons* |
| 434.00 | *acre-inches* |
| 10.33 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 9,762,018.00 | *gallons* |
| 359.50 | *acre-inches* |
| 8.56 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

CW-18S - 10/13/2022: Wheat, silage, soft dough

Field name: CW-18S  Crop: Wheat, silage, soft dough  Plant date: 10/13/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 230.97 | 21.21 | 449.34 | 1,993.26 |
| Fresh water | 0.00 | 0.00 | 0.00 | 327.17 |
| Atmospheric deposition | 14.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 244.97 | 21.21 | 449.34 | 2,320.43 |
| Anticipated crop nutrient removal | 198.00 | 30.60 | 149.40 | 0.00 |
| Actual crop nutrient removal | 191.62 | 42.36 | 379.20 | 792.69 |
| Nutrient balance | 53.36 | -21.15 | 70.14 | 1,527.75 |
| Applied to removed ratio | 1.28 | 0.50 | 1.18 | 2.93 |

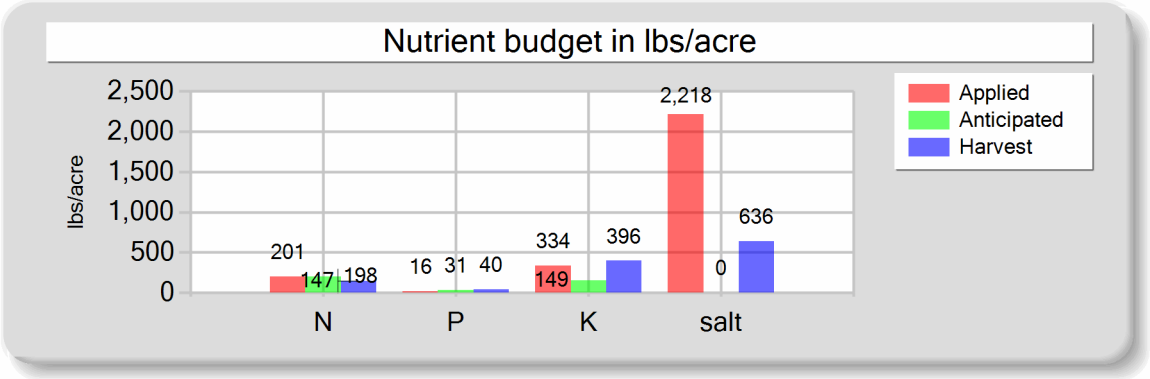
| Fresh water applied | |
|---|---|
| 12,415,000.00 | *gallons* |
| 457.20 | *acre-inches* |
| 12.03 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 13,039,968.00 | *gallons* |
| 480.22 | *acre-inches* |
| 12.64 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

CW-19N - 10/22/2022: Wheat, silage, soft dough

Field name: CW-19N  Crop: Wheat, silage, soft dough  Plant date: 10/22/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 170.07 | 16.17 | 333.73 | 1,575.42 |
| Fresh water | 17.08 | 0.00 | 0.00 | 642.45 |
| Atmospheric deposition | 14.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 201.15 | 16.17 | 333.73 | 2,217.87 |
| Anticipated crop nutrient removal | 198.00 | 30.60 | 149.40 | 0.00 |
| Actual crop nutrient removal | 146.83 | 39.77 | 396.12 | 636.24 |
| Nutrient balance | 54.32 | -23.60 | -62.40 | 1,581.62 |
| Applied to removed ratio | 1.37 | 0.41 | 0.84 | 3.49 |

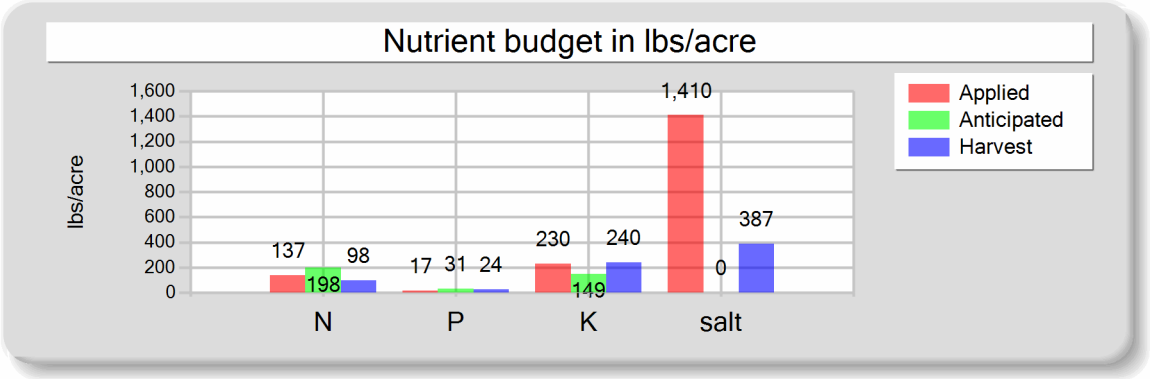
| Fresh water applied | |
|---|---|
| 10,755,500.00 | *gallons* |
| 396.09 | *acre-inches* |
| 5.66 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 19,316,904.00 | *gallons* |
| 711.38 | *acre-inches* |
| 10.16 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## CW-19S - 10/21/2022: Wheat, silage, soft dough

Field name: CW-19S  Crop: Wheat, silage, soft dough  Plant date: 10/21/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 129.60 | 17.40 | 229.68 | 1,201.98 |
| Fresh water | 0.36 | 0.00 | 0.00 | 208.45 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 136.96 | 17.40 | 229.68 | 1,410.43 |
| Anticipated crop nutrient removal | 198.00 | 30.60 | 149.40 | 0.00 |
| Actual crop nutrient removal | 97.60 | 24.40 | 240.38 | 386.77 |
| Nutrient balance | 39.36 | -7.00 | -10.69 | 1,023.67 |
| Applied to removed ratio | 1.40 | 0.71 | 0.96 | 3.65 |

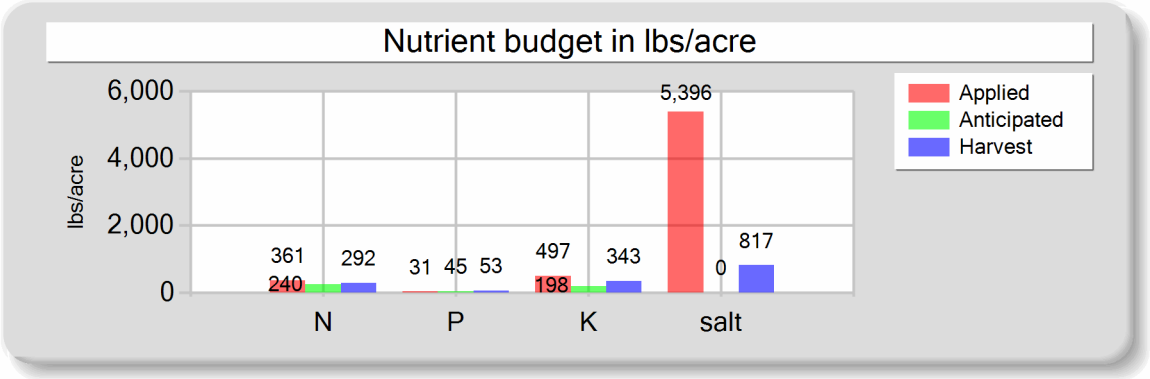
| Fresh water applied | |
|---|---|
| 8,716,200.00 | *gallons* |
| 320.99 | *acre-inches* |
| 5.84 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 11,905,802.00 | *gallons* |
| 438.45 | *acre-inches* |
| 7.97 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## CW-19S - 06/14/2023: Corn, silage

Field name: CW-19S  Crop: Corn, silage  Plant date: 06/14/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 52.08 | 9.30 | 6.15 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 213.13 | 21.95 | 490.97 | 2,393.24 |
| Fresh water | 88.41 | 0.00 | 0.00 | 3,003.18 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 360.62 | 31.25 | 497.12 | 5,396.41 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 291.87 | 53.30 | 342.63 | 817.24 |
| Nutrient balance | 68.75 | -22.05 | 154.49 | 4,579.18 |
| Applied to removed ratio | 1.24 | 0.59 | 1.45 | 6.60 |

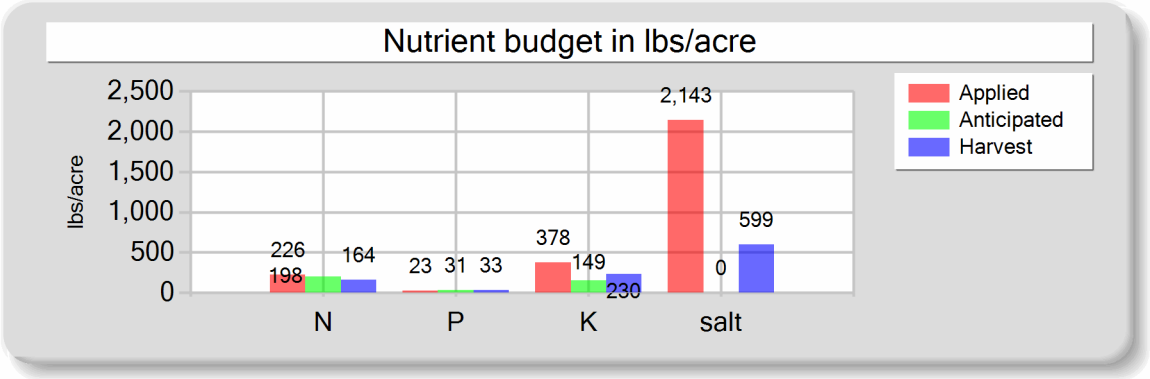
| Fresh water applied | |
|---|---|
| 32,162,360.00 | *gallons* |
| 1,184.43 | *acre-inches* |
| 21.54 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 13,094,460.00 | *gallons* |
| 482.22 | *acre-inches* |
| 8.77 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## CW-20 - 10/21/2022: Wheat, silage, soft dough

Field name: CW-20 Crop: Wheat, silage, soft dough Plant date: 10/21/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 215.15 | 23.17 | 377.79 | 1,754.06 |
| Fresh water | 4.11 | 0.00 | 0.00 | 388.71 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 226.27 | 23.17 | 377.79 | 2,142.77 |
| Anticipated crop nutrient removal | 198.00 | 30.60 | 149.40 | 0.00 |
| Actual crop nutrient removal | 163.67 | 33.37 | 230.41 | 599.05 |
| Nutrient balance | 62.60 | -10.20 | 147.39 | 1,543.71 |
| Applied to removed ratio | 1.38 | 0.69 | 1.64 | 3.58 |

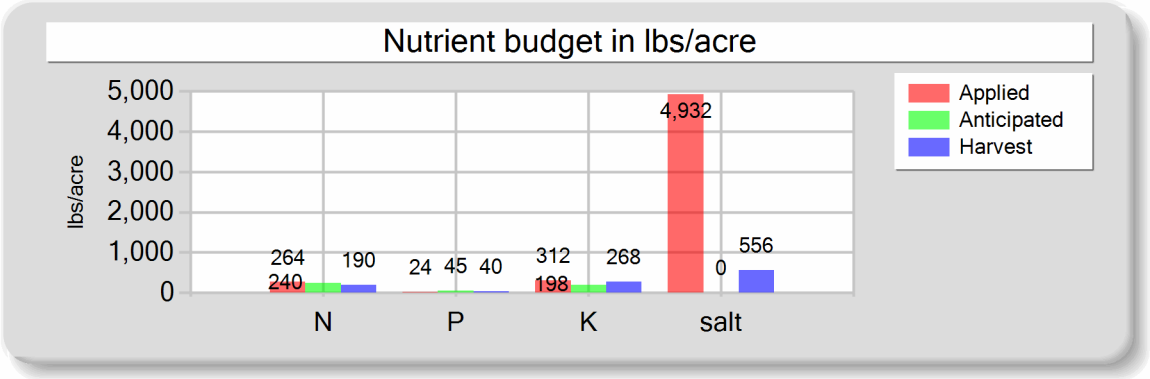
| Fresh water applied | |
|---|---|
| 7,170,000.00 | *gallons* |
| 264.05 | *acre-inches* |
| 7.33 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 11,034,330.00 | *gallons* |
| 406.36 | *acre-inches* |
| 11.29 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## CW-20 - 06/14/2023: Corn, silage

Field name: CW-20 Crop: Corn, silage Plant date: 06/14/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 52.08 | 9.30 | 6.15 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 113.98 | 14.49 | 305.37 | 1,857.00 |
| Fresh water | 91.37 | 0.00 | 0.00 | 3,075.35 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 264.43 | 23.79 | 311.52 | 4,932.35 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 190.42 | 39.82 | 268.32 | 555.69 |
| Nutrient balance | 74.01 | -16.03 | 43.20 | 4,376.66 |
| Applied to removed ratio | 1.39 | 0.60 | 1.16 | 8.88 |

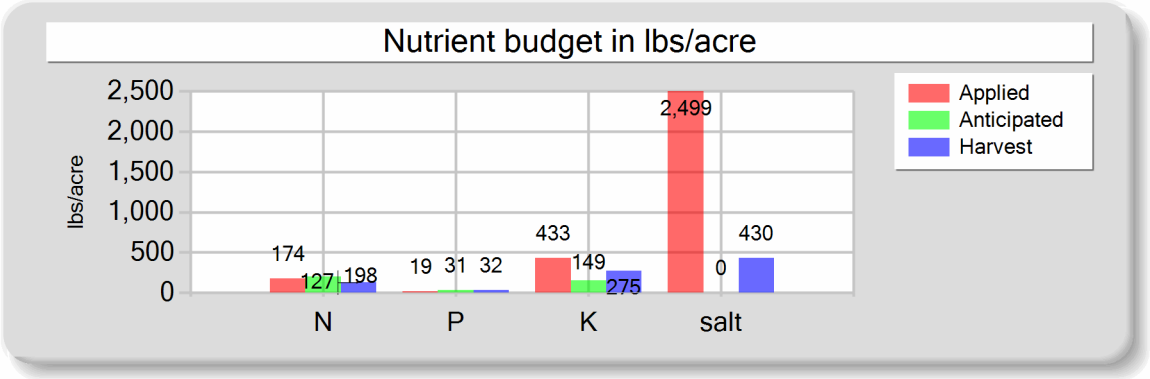
| Fresh water applied | |
|---|---|
| 26,230,080.00 | *gallons* |
| 965.96 | *acre-inches* |
| 26.83 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 5,894,562.00 | *gallons* |
| 217.08 | *acre-inches* |
| 6.03 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## CW-21 - 10/22/2022: Wheat, silage, soft dough

Field name: CW-21 Crop: Wheat, silage, soft dough Plant date: 10/22/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 166.56 | 19.39 | 432.79 | 2,360.39 |
| Fresh water | 0.00 | 0.00 | 0.00 | 138.44 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 173.56 | 19.39 | 432.79 | 2,498.83 |
| Anticipated crop nutrient removal | 198.00 | 30.60 | 149.40 | 0.00 |
| Actual crop nutrient removal | 126.83 | 31.95 | 274.95 | 429.85 |
| Nutrient balance | 46.73 | -12.56 | 157.83 | 2,068.97 |
| Applied to removed ratio | 1.37 | 0.61 | 1.57 | 5.81 |

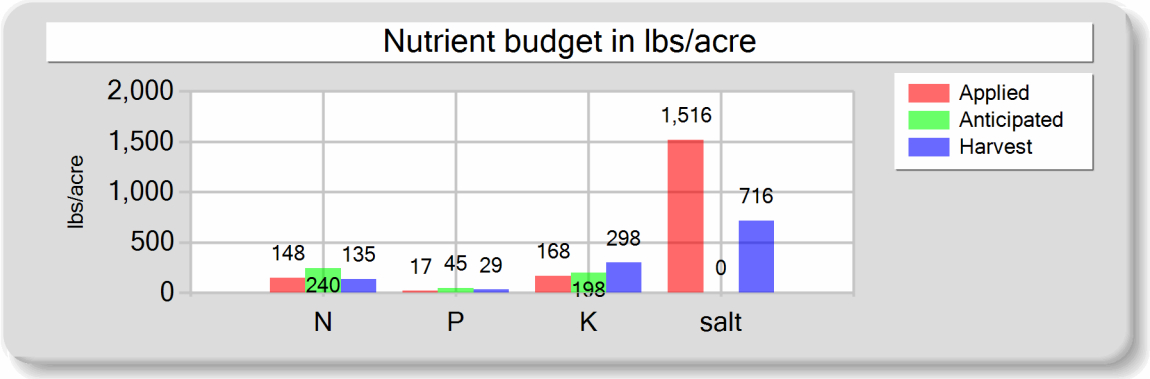
| Fresh water applied | |
|---|---|
| 8,433,000.00 | *gallons* |
| 310.56 | *acre-inches* |
| 5.09 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 10,731,162.00 | *gallons* |
| 395.19 | *acre-inches* |
| 6.48 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## CW-21 - 06/16/2023: Corn, silage

Field name: CW-21 Crop: Corn, silage Plant date: 06/16/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 52.08 | 9.30 | 6.15 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 84.98 | 7.22 | 162.32 | 620.37 |
| Fresh water | 3.73 | 0.00 | 0.00 | 895.52 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 147.80 | 16.52 | 168.47 | 1,515.89 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 135.47 | 29.03 | 298.03 | 716.04 |
| Nutrient balance | 12.33 | -12.51 | -129.56 | 799.85 |
| Applied to removed ratio | 1.09 | 0.57 | 0.57 | 2.12 |

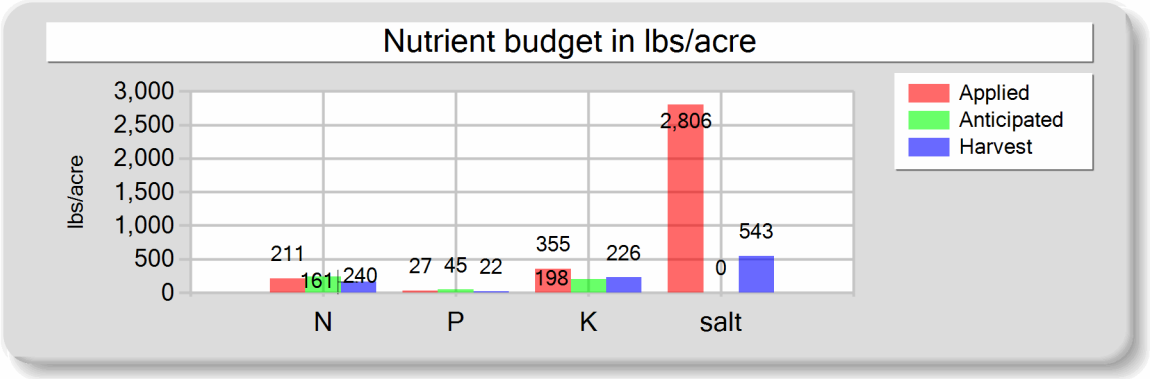
| Fresh water applied | |
|---|---|
| 43,665,500.00 | *gallons* |
| 1,608.05 | *acre-inches* |
| 26.36 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 6,211,992.00 | *gallons* |
| 228.77 | *acre-inches* |
| 3.75 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## CW-22 - 05/16/2023: Corn, silage

Field name: CW-22 Crop: Corn, silage Plant date: 05/16/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 52.08 | 9.30 | 6.15 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 145.02 | 17.55 | 348.74 | 2,224.18 |
| Fresh water | 0.00 | 0.00 | 0.00 | 582.16 |
| Atmospheric deposition | 14.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 211.10 | 26.85 | 354.89 | 2,806.34 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 161.49 | 22.02 | 226.08 | 543.19 |
| Nutrient balance | 49.61 | 4.83 | 128.80 | 2,263.16 |
| Applied to removed ratio | 1.31 | 1.22 | 1.57 | 5.17 |

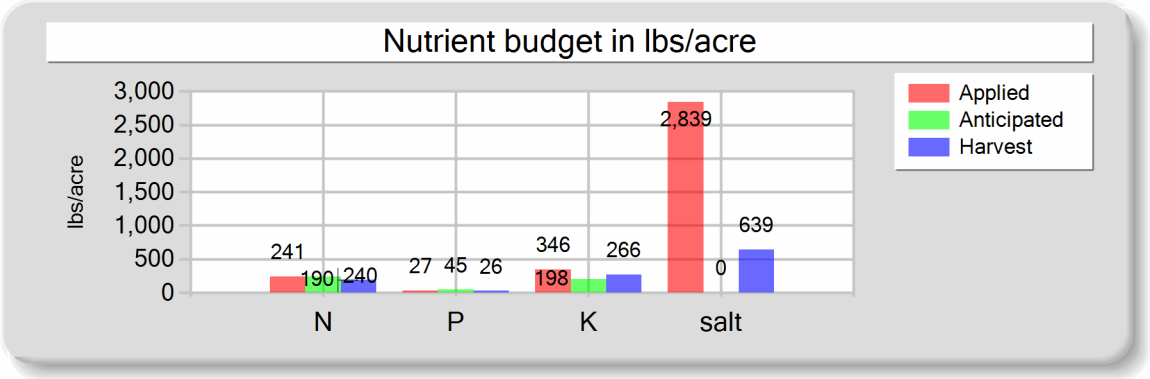
| Fresh water applied | |
|---|---|
| 30,230,000.00 | *gallons* |
| 1,113.27 | *acre-inches* |
| 21.41 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 14,020,812.00 | *gallons* |
| 516.34 | *acre-inches* |
| 9.93 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## CW-23 - 05/19/2023: Corn, silage

Field name: CW-23 Crop: Corn, silage Plant date: 05/19/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 52.08 | 9.30 | 6.15 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 168.27 | 17.78 | 339.49 | 1,907.55 |
| Fresh water | 7.06 | 0.00 | 0.00 | 931.02 |
| Atmospheric deposition | 14.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 241.41 | 27.08 | 345.64 | 2,838.57 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 190.00 | 25.91 | 266.00 | 639.10 |
| Nutrient balance | 51.41 | 1.17 | 79.64 | 2,199.48 |
| Applied to removed ratio | 1.27 | 1.05 | 1.30 | 4.44 |

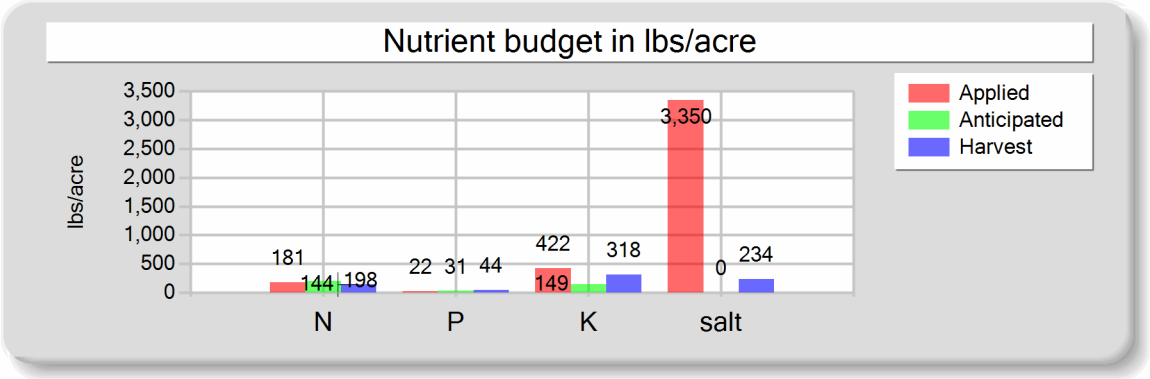
| Fresh water applied | |
|---|---|
| 21,333,000.00 | *gallons* |
| 785.62 | *acre-inches* |
| 21.82 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 12,542,802.00 | *gallons* |
| 461.91 | *acre-inches* |
| 12.83 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## CW-29 - 10/17/2022: Wheat, silage, soft dough

Field name: CW-29 Crop: Wheat, silage, soft dough Plant date: 10/17/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 167.50 | 21.67 | 422.24 | 2,987.29 |
| Fresh water | 0.00 | 0.00 | 0.00 | 362.69 |
| Atmospheric deposition | 14.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 181.50 | 21.67 | 422.24 | 3,349.99 |
| Anticipated crop nutrient removal | 198.00 | 30.60 | 149.40 | 0.00 |
| Actual crop nutrient removal | 143.91 | 43.97 | 317.81 | 233.86 |
| Nutrient balance | 37.59 | -22.30 | 104.43 | 3,116.13 |
| Applied to removed ratio | 1.26 | 0.49 | 1.33 | 14.32 |

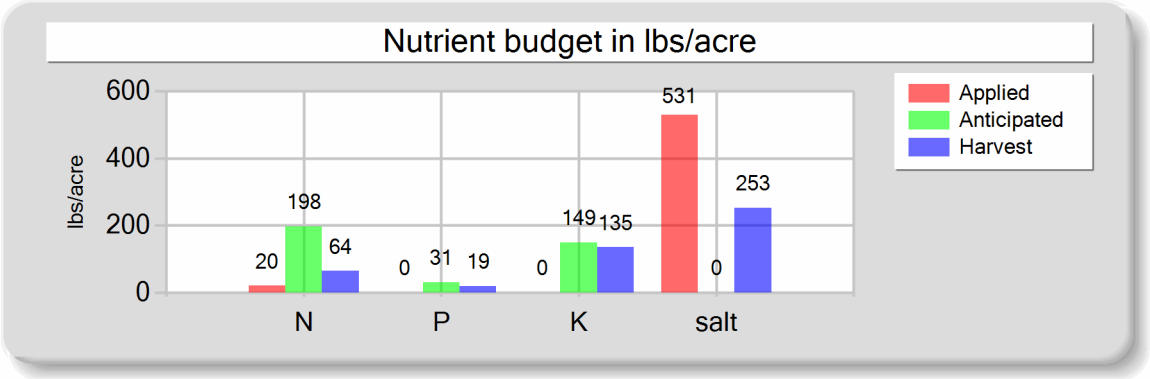
| Fresh water applied | |
|---|---|
| 28,975,000.00 | *gallons* |
| 1,067.05 | *acre-inches* |
| 13.34 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 26,539,986.00 | *gallons* |
| 977.38 | *acre-inches* |
| 12.22 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## CW-30 - 11/26/2022: Wheat, silage, soft dough

Field name: CW-30 Crop: Wheat, silage, soft dough Plant date: 11/26/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 13.45 | 0.00 | 0.00 | 530.70 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 20.45 | 0.00 | 0.00 | 530.70 |
| Anticipated crop nutrient removal | 198.00 | 30.60 | 149.40 | 0.00 |
| Actual crop nutrient removal | 64.38 | 19.19 | 134.96 | 252.58 |
| Nutrient balance | -43.93 | -19.19 | -134.96 | 278.12 |
| Applied to removed ratio | 0.32 | 0.00 | 0.00 | 2.10 |

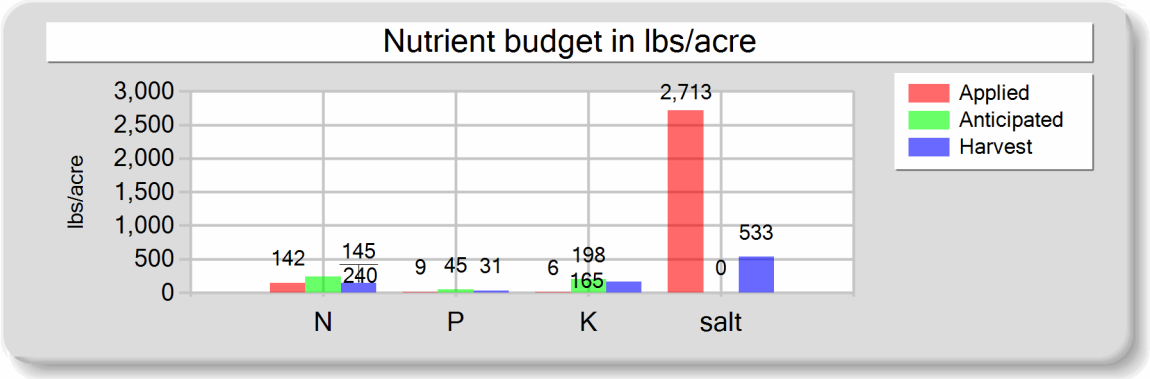
| Fresh water applied | |
|---|---|
| 9,221,380.00 | *gallons* |
| 339.59 | *acre-inches* |
| 8.28 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 0.00 | *gallons* |
| 0.00 | *acre-inches* |
| 0.00 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## CW-30 - 06/19/2023: Corn, silage

Field name: CW-30 Crop: Corn, silage Plant date: 06/19/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 92.08 | 9.30 | 6.15 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 43.39 | 0.00 | 0.00 | 2,712.90 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 142.47 | 9.30 | 6.15 | 2,712.90 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 144.67 | 30.78 | 164.68 | 532.51 |
| Nutrient balance | -2.20 | -21.48 | -158.53 | 2,180.39 |
| Applied to removed ratio | 0.98 | 0.30 | 0.04 | 5.09 |

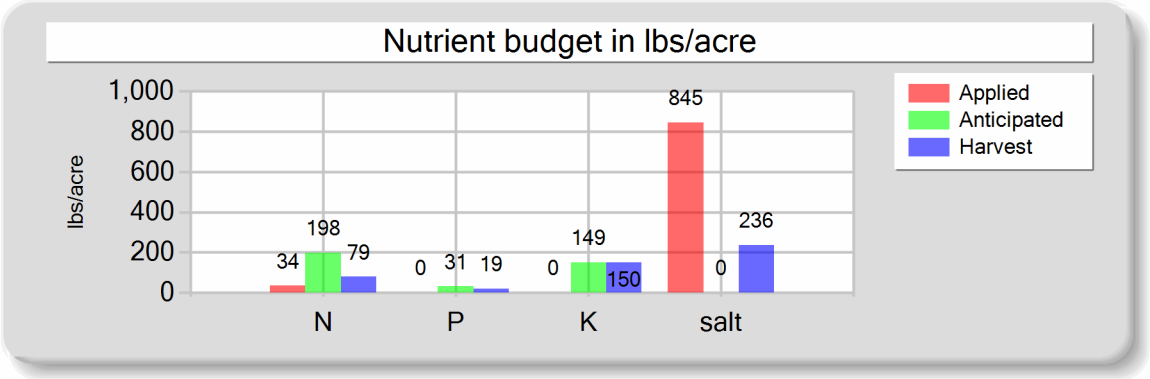
| Fresh water applied | |
|---|---|
| 32,943,500.00 | *gallons* |
| 1,213.20 | *acre-inches* |
| 29.59 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 0.00 | *gallons* |
| 0.00 | *acre-inches* |
| 0.00 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

CW-31 - 11/26/2022: Wheat, silage, soft dough

Field name: CW-31 Crop: Wheat, silage, soft dough Plant date: 11/26/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 27.03 | 0.00 | 0.00 | 845.36 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 34.03 | 0.00 | 0.00 | 845.36 |
| Anticipated crop nutrient removal | 198.00 | 30.60 | 149.40 | 0.00 |
| Actual crop nutrient removal | 78.55 | 18.59 | 149.91 | 235.66 |
| Nutrient balance | -44.52 | -18.59 | -149.91 | 609.70 |
| Applied to removed ratio | 0.43 | 0.00 | 0.00 | 3.59 |

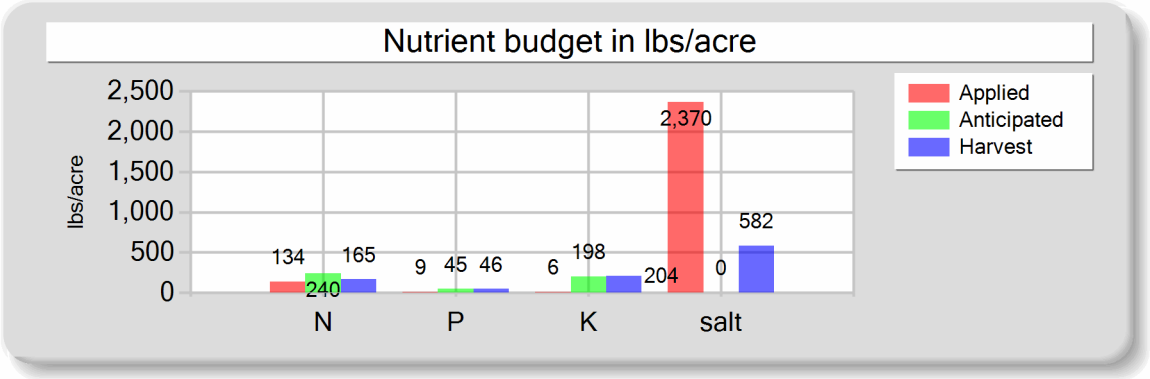
| Fresh water applied | |
|---|---|
| 8,088,870.00 | *gallons* |
| 297.89 | *acre-inches* |
| 8.51 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 0.00 | *gallons* |
| 0.00 | *acre-inches* |
| 0.00 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## CW-31 - 06/20/2023: Corn, silage

Field name: CW-31 Crop: Corn, silage Plant date: 06/20/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 92.08 | 9.30 | 6.15 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 35.08 | 0.00 | 0.00 | 2,370.24 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 134.16 | 9.30 | 6.15 | 2,370.24 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 164.67 | 45.84 | 203.72 | 582.29 |
| Nutrient balance | -30.51 | -36.54 | -197.57 | 1,787.94 |
| Applied to removed ratio | 0.81 | 0.20 | 0.03 | 4.07 |

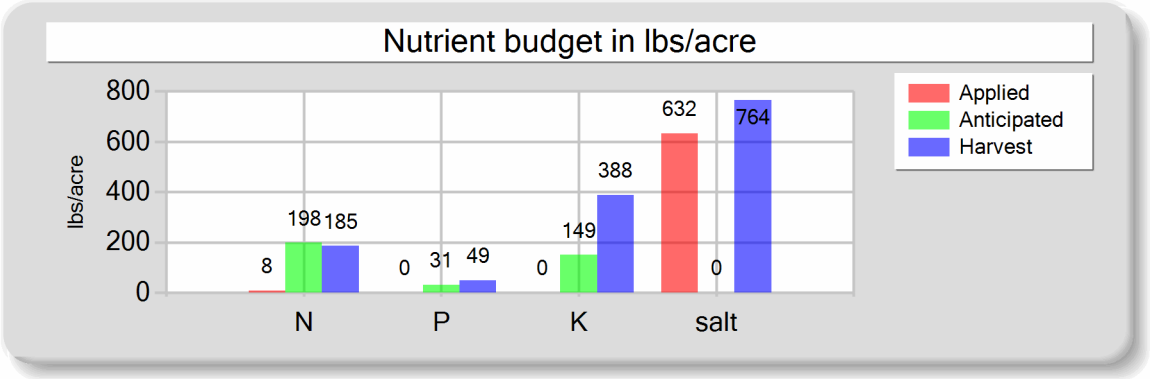
| Fresh water applied | |
|---|---|
| 28,551,500.00 | *gallons* |
| 1,051.45 | *acre-inches* |
| 30.04 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 0.00 | *gallons* |
| 0.00 | *acre-inches* |
| 0.00 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## CW-9 - 10/27/2022: Wheat, silage, soft dough

Field name: CW-9 Crop: Wheat, silage, soft dough Plant date: 10/27/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 1.40 | 0.00 | 0.00 | 632.50 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 8.40 | 0.00 | 0.00 | 632.50 |
| Anticipated crop nutrient removal | 198.00 | 30.60 | 149.40 | 0.00 |
| Actual crop nutrient removal | 185.09 | 49.23 | 387.90 | 763.99 |
| Nutrient balance | -176.69 | -49.23 | -387.90 | -131.49 |
| Applied to removed ratio | 0.05 | 0.00 | 0.00 | 0.83 |

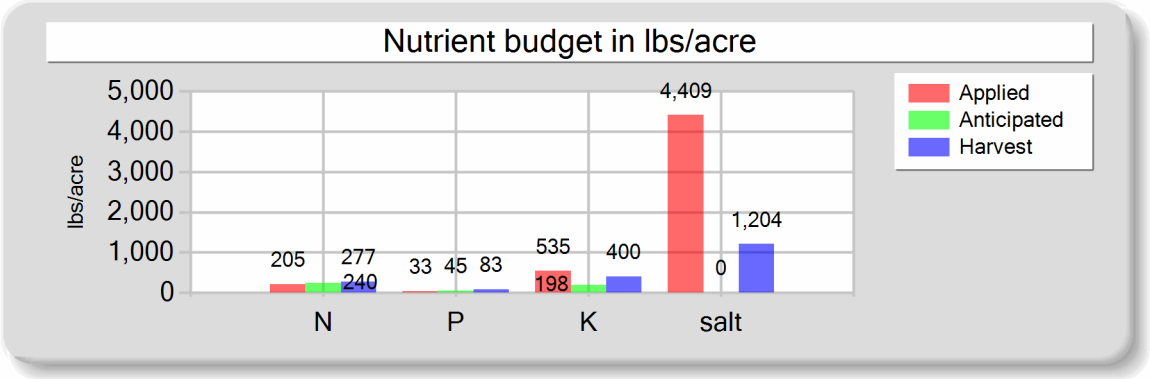
| Fresh water applied | |
|---|---|
| 6,544,000.00 | *gallons* |
| 240.99 | *acre-inches* |
| 16.07 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 0.00 | *gallons* |
| 0.00 | *acre-inches* |
| 0.00 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## CW-9 - 06/14/2023: Corn, silage

Field name: CW-9 Crop: Corn, silage Plant date: 06/14/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 52.08 | 9.30 | 6.15 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 145.75 | 23.81 | 528.61 | 3,561.73 |
| Fresh water | 0.00 | 0.00 | 0.00 | 846.92 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 204.83 | 33.11 | 534.76 | 4,408.65 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 277.42 | 82.94 | 400.41 | 1,204.08 |
| Nutrient balance | -72.59 | -49.83 | 134.36 | 3,204.58 |
| Applied to removed ratio | 0.74 | 0.40 | 1.34 | 3.66 |

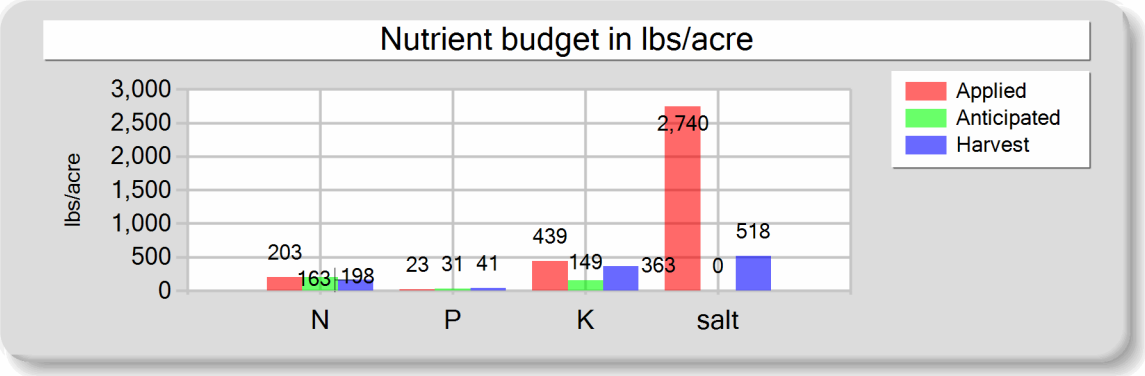
| Fresh water applied | |
|---|---|
| 13,182,500.00 | *gallons* |
| 485.47 | *acre-inches* |
| 32.36 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 1,845,000.00 | *gallons* |
| 67.95 | *acre-inches* |
| 4.53 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

LP-1 - 10/10/2022: Wheat, silage, soft dough

Field name: LP-1 Crop: Wheat, silage, soft dough Plant date: 10/10/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 194.87 | 22.51 | 439.29 | 2,634.52 |
| Fresh water | 1.59 | 0.00 | 0.00 | 105.35 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 203.46 | 22.51 | 439.29 | 2,739.87 |
| Anticipated crop nutrient removal | 198.00 | 30.60 | 149.40 | 0.00 |
| Actual crop nutrient removal | 162.98 | 41.05 | 363.27 | 517.54 |
| Nutrient balance | 40.48 | -18.54 | 76.02 | 2,222.33 |
| Applied to removed ratio | 1.25 | 0.55 | 1.21 | 5.29 |

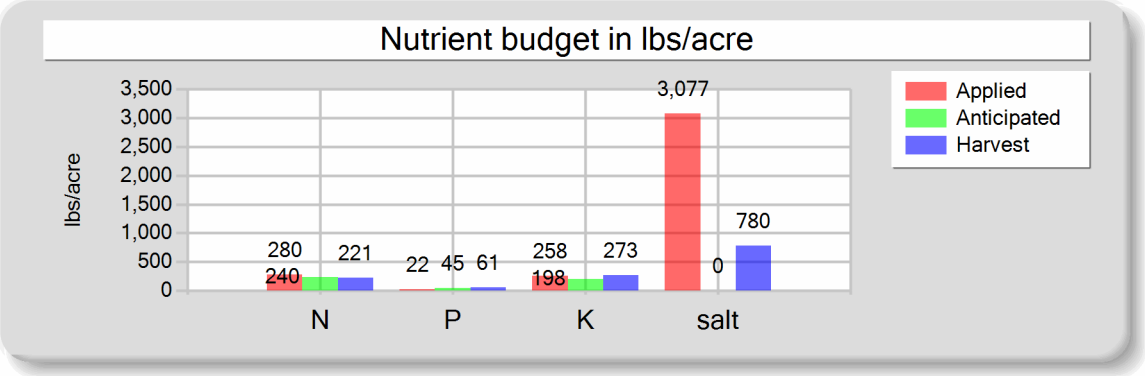
| Fresh water applied | |
|---|---|
| 1,903,840.00 | *gallons* |
| 70.11 | *acre-inches* |
| 1.08 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 24,917,964.00 | *gallons* |
| 917.64 | *acre-inches* |
| 14.12 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

LP-1 - 07/09/2023: Corn, silage

Field name: LP-1 Crop: Corn, silage Plant date: 07/09/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 92.08 | 9.30 | 6.15 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 126.74 | 12.69 | 252.07 | 1,300.38 |
| Fresh water | 54.04 | 0.00 | 0.00 | 1,776.64 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 279.86 | 21.99 | 258.22 | 3,077.01 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 220.67 | 61.42 | 272.99 | 780.29 |
| Nutrient balance | 59.19 | -39.43 | -14.77 | 2,296.72 |
| Applied to removed ratio | 1.27 | 0.36 | 0.95 | 3.94 |

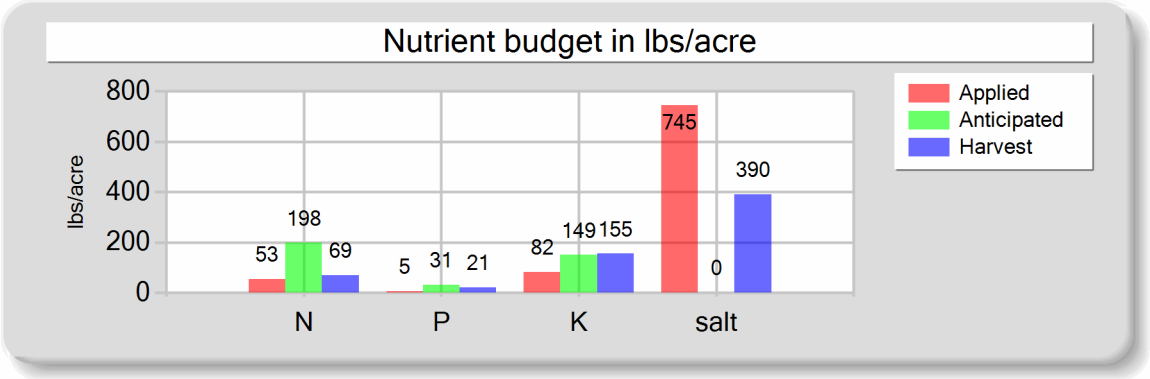
| Fresh water applied | |
|---|---|
| 42,941,830.00 | *gallons* |
| 1,581.40 | *acre-inches* |
| 24.33 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 15,141,792.00 | *gallons* |
| 557.62 | *acre-inches* |
| 8.58 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

LP-2 - 10/11/2022: Wheat, silage, soft dough

Field name: LP-2 Crop: Wheat, silage, soft dough Plant date: 10/11/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 39.43 | 4.61 | 81.88 | 534.43 |
| Fresh water | 0.00 | 0.00 | 0.00 | 210.13 |
| Atmospheric deposition | 14.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 53.43 | 4.61 | 81.88 | 744.56 |
| Anticipated crop nutrient removal | 198.00 | 30.60 | 149.40 | 0.00 |
| Actual crop nutrient removal | 68.54 | 20.56 | 154.71 | 389.72 |
| Nutrient balance | -15.11 | -15.95 | -72.83 | 354.84 |
| Applied to removed ratio | 0.78 | 0.22 | 0.53 | 1.91 |

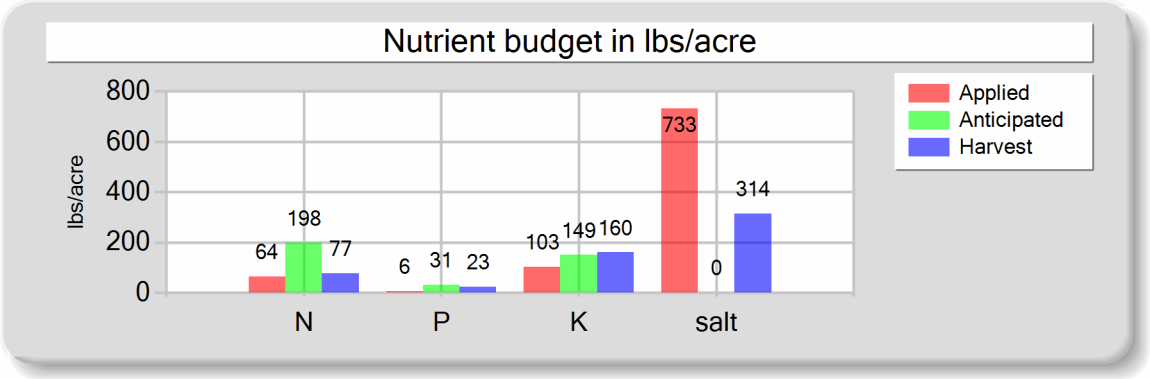
| Fresh water applied | |
|---|---|
| 9,442,500.00 | *gallons* |
| 347.74 | *acre-inches* |
| 7.73 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 4,524,186.00 | *gallons* |
| 166.61 | *acre-inches* |
| 3.70 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## LP-3 - 10/11/2022: Wheat, silage, soft dough

Field name: LP-3 Crop: Wheat, silage, soft dough Plant date: 10/11/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 49.79 | 5.83 | 103.39 | 674.76 |
| Fresh water | 0.00 | 0.00 | 0.00 | 58.10 |
| Atmospheric deposition | 14.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 63.79 | 5.83 | 103.39 | 732.86 |
| Anticipated crop nutrient removal | 198.00 | 30.60 | 149.40 | 0.00 |
| Actual crop nutrient removal | 77.48 | 23.48 | 159.65 | 313.82 |
| Nutrient balance | -13.69 | -17.65 | -56.26 | 419.05 |
| Applied to removed ratio | 0.82 | 0.25 | 0.65 | 2.34 |

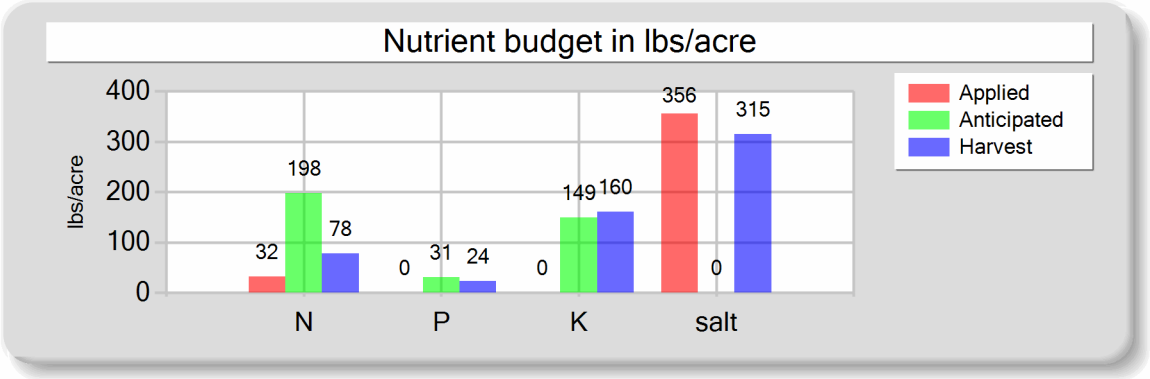
| Fresh water applied | |
|---|---|
| 1,334,500.00 | *gallons* |
| 49.15 | *acre-inches* |
| 2.14 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 2,919,528.00 | *gallons* |
| 107.52 | *acre-inches* |
| 4.67 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## LP-4 - 10/12/2022: Wheat, silage, soft dough

Field name: LP-4 Crop: Wheat, silage, soft dough Plant date: 10/12/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 17.73 | 0.00 | 0.00 | 355.84 |
| Atmospheric deposition | 14.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 31.73 | 0.00 | 0.00 | 355.84 |
| Anticipated crop nutrient removal | 198.00 | 30.60 | 149.40 | 0.00 |
| Actual crop nutrient removal | 77.81 | 23.58 | 160.34 | 315.19 |
| Nutrient balance | -46.09 | -23.58 | -160.34 | 40.65 |
| Applied to removed ratio | 0.41 | 0.00 | 0.00 | 1.13 |

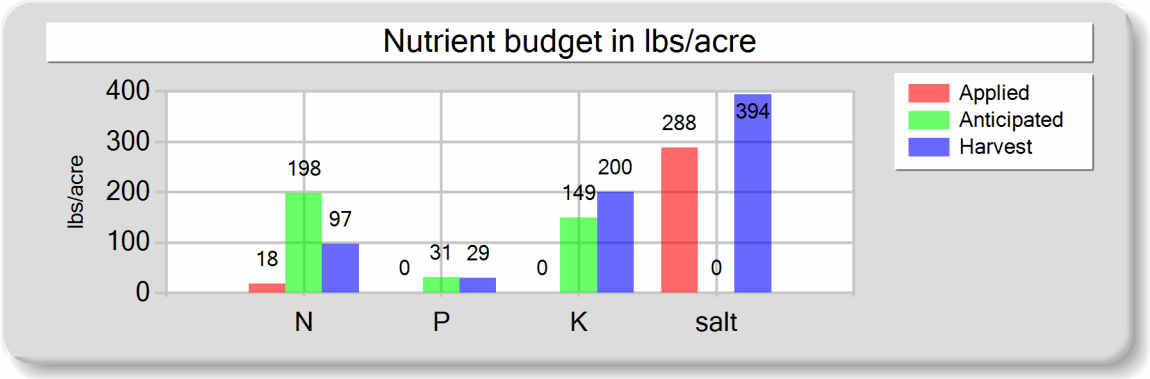
| Fresh water applied | |
|---|---|
| 391,200.00 | *gallons* |
| 14.41 | *acre-inches* |
| 1.44 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 0.00 | *gallons* |
| 0.00 | *acre-inches* |
| 0.00 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

LP-5 - 10/12/2022: Wheat, silage, soft dough

Field name: LP-5 Crop: Wheat, silage, soft dough Plant date: 10/12/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 4.34 | 0.00 | 0.00 | 287.96 |
| Atmospheric deposition | 14.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 18.34 | 0.00 | 0.00 | 287.96 |
| Anticipated crop nutrient removal | 198.00 | 30.60 | 149.40 | 0.00 |
| Actual crop nutrient removal | 97.27 | 29.48 | 200.43 | 393.98 |
| Nutrient balance | -78.92 | -29.48 | -200.43 | -106.02 |
| Applied to removed ratio | 0.19 | 0.00 | 0.00 | 0.73 |

| Fresh water applied | |
|---|---|
| 640,500.00 | *gallons* |
| 23.59 | *acre-inches* |
| 2.95 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 0.00 | *gallons* |
| 0.00 | *acre-inches* |
| 0.00 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

NUTRIENT ANALYSES

## A. MANURE ANALYSES

*No manure analyses entered.*

## B. PROCESS WASTEWATER ANALYSES

### Lagoon Waste Water - Cannery (Lab Number: L69818-04)

Sample and source description: Lagoon Waste Water - Cannery (Lab Number: L69818-04)

Sample date: 10/18/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: _____

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 100.00 | 3.20 | | 0.00 | 8.50 | 191.00 | | | | | | | | 1,100.00 | 730 |
| **DL** | 10.00 | 2.00 | | 2.00 | 0.20 | 0.50 | | | | | | | | 100.00 | 10 |

### Lagoon Waste Water - CW-WSP #1 (Lab Number: L69818-01)

Sample and source description: Lagoon Waste Water - CW-WSP #1 (Lab Number: L69818-01)

Sample date: 10/18/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: _____

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 142.00 | 24.40 | | 0.00 | 23.20 | 515.00 | | | | | | | | 5,220.00 | 3,470 |
| **DL** | 10.00 | 2.00 | | 2.00 | 0.20 | 0.50 | | | | | | | | 100.00 | 10 |

### Lagoon Waste Water - North Pond (Lab Number: L69818-02)

Sample and source description: Lagoon Waste Water - North Pond (Lab Number: L69818-02)

Sample date: 10/18/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: _____

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 50.00 | 2.70 | | 0.00 | 3.90 | 88.00 | | | | | | | | 820.00 | 544 |
| **DL** | 10.00 | 2.00 | | 2.00 | 0.20 | 0.50 | | | | | | | | 100.00 | 10 |

## Lagoon Waste Water - Peck Drain (Lab Number: L69818-05)

Sample and source description: Lagoon Waste Water - Peck Drain (Lab Number: L69818-05)

Sample date: 10/18/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 86.00 | 0.00 | | 0.00 | 3.90 | 87.50 | | | | | | | | 180.00 | 120 |
| **DL** | 10.00 | 2.00 | | 2.00 | 0.20 | 0.50 | | | | | | | | 100.00 | 10 |

## Lagoon Waste Water - South Pond (Lab Number: L69818-03)

Sample and source description: Lagoon Waste Water - South Pond (Lab Number: L69818-03)

Sample date: 10/18/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 35.00 | 31.70 | | 0.00 | 32.40 | 290.00 | | | | | | | | 3,350.00 | 2,220 |
| **DL** | 10.00 | 2.00 | | 2.00 | 0.20 | 0.50 | | | | | | | | 100.00 | 10 |

## Lagoon Waste Water - Cannery (Lab Number: L72896-04)

Sample and source description: Lagoon Waste Water - Cannery (Lab Number: L72896-04)

Sample date: 11/14/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 47.00 | 0.00 | | 0.00 | 5.50 | 97.60 | | | | | | | | 960.00 | 637 |
| **DL** | 10.00 | 2.00 | | 2.00 | 0.20 | 0.50 | | | | | | | | 100.00 | 10 |

## Lagoon Waste Water - CW-WSP #1 (Lab Number: L72896-01)

Sample and source description: Lagoon Waste Water - CW-WSP #1 (Lab Number: L72896-01)

Sample date: 11/14/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 205.00 | 201.00 | | 0.00 | 35.30 | 259.00 | | | | | | | | 4,270.00 | 2,840 |
| **DL** | 10.00 | 2.00 | | 2.00 | 0.20 | 0.50 | | | | | | | | 100.00 | 10 |

## Lagoon Waste Water - North Pond (Lab Number: L72896-02)

Sample and source description: Lagoon Waste Water - North Pond (Lab Number: L72896-02)

Sample date: 11/14/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 79.00 | 0.00 | | 0.00 | 0.00 | 43.30 | | | | | | | | 1,000.00 | 664 |
| **DL** | 10.00 | 2.00 | | 2.00 | 0.20 | 0.50 | | | | | | | | 100.00 | 10 |

## Lagoon Waste Water - Peck Drain (Lab Number: L72896-03)

Sample and source description: Lagoon Waste Water - Peck Drain (Lab Number: L72896-03)

Sample date: 11/14/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 10.00 | 0.00 | | 0.00 | 0.00 | 54.60 | | | | | | | | 710.00 | 471 |
| **DL** | 10.00 | 2.00 | | 2.00 | 0.20 | 0.50 | | | | | | | | 100.00 | 10 |

## Lagoon Waste Water - Cannery (Lab Number: L73906-03)

Sample and source description: Lagoon Waste Water - Cannery (Lab Number: L73906-03)

Sample date: 12/04/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 128.00 | 0.00 | | 0.00 | 6.70 | 146.00 | | | | | | | | 930.00 | 618 |
| **DL** | 10.00 | 2.00 | | 2.00 | 0.20 | 0.50 | | | | | | | | 100.00 | 10 |

## Lagoon Waste Water - CW-WSP #1 (Lab Number: L73906-01)

Sample and source description: Lagoon Waste Water - CW-WSP #1 (Lab Number: L73906-01)

Sample date: 12/04/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 234.00 | 230.00 | | 0.00 | 40.20 | 278.00 | | | | | | | | 4,380.00 | 2,910 |
| **DL** | 10.00 | 2.00 | | 2.00 | 0.20 | 0.50 | | | | | | | | 100.00 | 10 |

Lagoon Waste Water - North Pond (Lab Number: L73906-02)

Sample and source description: Lagoon Waste Water - North Pond (Lab Number: L73906-02)

Sample date: 12/04/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 110.00 | 0.00 | | 0.00 | 0.00 | 59.50 | | | | | | | | 920.00 | 611 |
| **DL** | 10.00 | 2.00 | | 2.00 | 0.20 | 0.50 | | | | | | | | 100.00 | 10 |

Lagoon Waste Water - Peck Drain (Lab Number: L73906-04)

Sample and source description: Lagoon Waste Water - Peck Drain (Lab Number: L73906-04)

Sample date: 12/04/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 71.00 | 0.00 | | 0.00 | 4.10 | 70.90 | | | | | | | | 670.00 | 445 |
| **DL** | 10.00 | 2.00 | | 2.00 | 0.20 | 0.50 | | | | | | | | 100.00 | 10 |

## C. FRESH WATER ANALYSES

MID

MID (2021)

Sample description: MID (2021)

Sample date: 10/13/2021 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 0.00 | | 0.00 | | | | | | | | 40.00 | 74 |
| **DL** | 1.40 | | 0.40 | | | | | | | | 1.00 | 20 |

Peck Drain

## Peck Drain

### Peck Drain (Report No. L69818-05)

Sample description: Peck Drain (Report No. L69818-05)

Sample date: 10/18/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | | 0.00 | 0.00 | | | | | | | | 180.00 | 120 |
| **DL** | | 2.00 | 2.00 | | | | | | | | 100.00 | 10 |

### Peck Drain (Report No. L72896-03)

Sample description: Peck Drain (Report No. L72896-03)

Sample date: 11/14/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | | 0.00 | 0.00 | | | | | | | | 710.00 | 471 |
| **DL** | | 2.00 | 2.00 | | | | | | | | 100.00 | 10 |

### Peck Drain (Report No. L73906-04)

Sample description: Peck Drain (Report No. L73906-04)

Sample date: 12/04/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | | 0.00 | 0.00 | | | | | | | | 670.00 | 445 |
| **DL** | | 2.00 | 2.00 | | | | | | | | 100.00 | 10 |

## Well 113.5

## Well 113.5

### Well 113.5 5-Year (2022 Report No. W41113-13)

Sample description: Well 113.5 5-Year (2022 Report No. W41113-13)

Sample date: 12/06/2022 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | | | 10.50 | 42.40 | 9.70 | 56.10 | 217.00 | | 15.50 | 17.30 | 520.00 | 358 |
| **DL** | | | 0.10 | 1.00 | 1.00 | 1.00 | 1.00 | | 1.00 | 1.00 | 10.00 | 10 |

## Well 130

### Well 130 5-Year (2022 Report No. W41113-07)

Sample description: Well 130 5-Year (2022 Report No. W41113-07)

Sample date: 12/06/2022 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | | | 54.30 | 142.00 | 46.00 | 134.00 | 475.00 | | 58.60 | 122.00 | 1,590.00 | 1,090 |
| **DL** | | | 0.10 | 1.00 | 1.00 | 1.00 | 1.00 | | 1.00 | 1.00 | 10.00 | 10 |

## Well 131

### Well 131 5-Year (2022 Report No. W41113-06)

Sample description: Well 131 5-Year (2022 Report No. W41113-06)

Sample date: 12/06/2022 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | | | 7.20 | 138.00 | 50.00 | 80.70 | 459.00 | | 177.00 | 34.00 | 1,270.00 | 874 |
| **DL** | | | 0.10 | 1.00 | 1.00 | 1.00 | 1.00 | | 1.00 | 1.00 | 10.00 | 10 |

## Well 132

## Well 132

### Well 132 5-Year (2022 Report No. W41113-05)

Sample description: Well 132 5-Year (2022 Report No. W41113-05)

Sample date: 12/06/2022 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | | | 17.90 | 119.00 | 35.90 | 97.30 | 495.00 | | 66.90 | 64.50 | 1,210.00 | 832 |
| **DL** | | | 0.10 | 1.00 | 1.00 | 1.00 | 1.00 | | 1.00 | 1.00 | 10.00 | 10 |

## Well 133

### Well 133 5-Year (2022 Report No. W41113-04)

Sample description: Well 133 5-Year (2022 Report No. W41113-04)

Sample date: 12/06/2022 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | | | 8.70 | 118.00 | 39.10 | 74.80 | 464.00 | | 98.60 | 40.10 | 1,130.00 | 777 |
| **DL** | | | 0.10 | 1.00 | 1.00 | 1.00 | 1.00 | | 1.00 | 1.00 | 10.00 | 10 |

## Well 134

### Well 134 5-Year (2022 Report No. W41113-03)

Sample description: Well 134 5-Year (2022 Report No. W41113-03)

Sample date: 12/06/2022 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | | | 11.00 | 118.00 | 38.00 | 100.00 | 437.00 | | 130.00 | 52.10 | 1,210.00 | 832 |
| **DL** | | | 0.10 | 1.00 | 1.00 | 1.00 | 1.00 | | 1.00 | 1.00 | 10.00 | 10 |

## Well 135

## Well 135

### Well 135 5-Year (2022 Report No. W41113-02)

Sample description: Well 135 5-Year (2022 Report No. W41113-02)

Sample date: 12/06/2022 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | | | 1.10 | 91.20 | 18.00 | 71.30 | 371.00 | | 60.50 | 41.80 | 850.00 | 585 |
| **DL** | | | 0.10 | 1.00 | 1.00 | 1.00 | 1.00 | | 1.00 | 1.00 | 10.00 | 10 |

## Well 136

### Well 136 5-Year (2022 Report No. W41113-08)

Sample description: Well 136 5-Year (2022 Report No. W41113-08)

Sample date: 12/06/2022 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | | | 6.50 | 46.80 | 9.50 | 74.50 | 258.00 | | 28.80 | 29.90 | 630.00 | 431 |
| **DL** | | | 0.10 | 1.00 | 1.00 | 1.00 | 1.00 | | 1.00 | 1.00 | 10.00 | 10 |

## Well 137

### Well 137 5-Year (2022 Report No. W41113-09)

Sample description: Well 137 5-Year (2022 Report No. W41113-09)

Sample date: 12/06/2022 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | | | 3.80 | 46.20 | 11.00 | 69.90 | 282.00 | | 23.10 | 20.40 | 590.00 | 404 |
| **DL** | | | 0.10 | 1.00 | 1.00 | 1.00 | 1.00 | | 1.00 | 1.00 | 10.00 | 10 |

## Well 139

## Well 139

### Well 139 5-Year (2022 Report No. W41113-10)

Sample description: Well 139 5-Year (2022 Report No. W41113-10)

Sample date: 12/06/2022 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | | | 0.70 | 7.30 | 1.20 | 59.60 | 137.00 | | 12.40 | 19.30 | 320.00 | 218 |
| **DL** | | | 0.10 | 1.00 | 1.00 | 1.00 | 1.00 | | 1.00 | 1.00 | 10.00 | 10 |

## Well 140

### Well 140 5-Year (2022 Report No. W41113-01)

Sample description: Well 140 5-Year (2022 Report No. W41113-01)

Sample date: 12/06/2022 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | | | 13.90 | 111.00 | 31.00 | 78.40 | 448.00 | | 49.00 | 65.60 | 1,050.00 | 722 |
| **DL** | | | 0.10 | 1.00 | 1.00 | 1.00 | 1.00 | | 1.00 | 1.00 | 10.00 | 10 |

## Well 144

### Well 144 5-Year (2022 Report No. W41113-12)

Sample description: Well 144 5-Year (2022 Report No. W41113-12)

Sample date: 12/06/2022 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | | | 8.10 | 108.00 | 32.60 | 86.00 | 461.00 | | 58.60 | 64.50 | 1,070.00 | 736 |
| **DL** | | | 0.10 | 1.00 | 1.00 | 1.00 | 1.00 | | 1.00 | 1.00 | 10.00 | 10 |

## Well 149

## Well 149

### Well 149 5-Year (2022 Report No. W41113-11)

Sample description: Well 149 5-Year (2022 Report No. W41113-11)

Sample date: 12/06/2022 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | | | 10.00 | 58.00 | 16.30 | 82.60 | 337.00 | | 28.10 | 25.80 | 720.00 | 496 |
| **DL** | | | 0.10 | 1.00 | 1.00 | 1.00 | 1.00 | | 1.00 | 1.00 | 10.00 | 10 |

## Well 80

### Well 80 5-Year (2022 Report No. W41114-02)

Sample description: Well 80 5-Year (2022 Report No. W41114-02)

Sample date: 12/06/2022 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | | | 31.60 | 97.80 | 26.90 | 40.70 | 254.00 | | 45.40 | 39.30 | 900.00 | 616 |
| **DL** | | | 0.10 | 1.00 | 1.00 | 1.00 | 1.00 | | 1.00 | 1.00 | 10.00 | 10 |

## Well 82

### Well 82 5-Year (2022 Report No. W41114-01)

Sample description: Well 82 5-Year (2022 Report No. W41114-01)

Sample date: 12/06/2022 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | | | 67.20 | 153.00 | 44.80 | 91.10 | 428.00 | | 40.60 | 86.90 | 1,440.00 | 991 |
| **DL** | | | 0.10 | 1.00 | 1.00 | 1.00 | 1.00 | | 1.00 | 1.00 | 10.00 | 10 |

## Well 91.5

Well 91.5

Well 91.5 5-Year (2022 Report No. W41113-15)

Sample description: Well 91.5 5-Year (2022 Report No. W41113-15)

Sample date: 12/06/2022 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | | | 34.70 | 120.00 | 35.00 | 147.00 | 476.00 | | 76.80 | 93.20 | 1,410.00 | 970 |
| **DL** | | | 0.10 | 1.00 | 1.00 | 1.00 | 1.00 | | 1.00 | 1.00 | 10.00 | 10 |

Well 92

Well 92 5-Year (2022 Report No. W41113-14)

Sample description: Well 92 5-Year (2022 Report No. W41113-14)

Sample date: 12/06/2022 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | | | 41.60 | 155.00 | 46.80 | 195.00 | 647.00 | | 84.20 | 149.00 | 1,910.00 | 1,310 |
| **DL** | | | 0.10 | 1.00 | 1.00 | 1.00 | 1.00 | | 1.00 | 1.00 | 10.00 | 10 |

## D. SOIL ANALYSES

*No soil analyses entered.*

## E. PLANT TISSUE ANALYSES

CW-12 - 10/28/2022: Wheat, silage, soft dough

## CW-12 - 10/28/2022: Wheat, silage, soft dough

### Sample H51001-02 Lab No. 2105

Sample and source description: Sample H51001-02 Lab No. 2105

Sample date: 05/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.4 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 11,500.00 | 3,200.00 | 23,400.00 | | 4.02 |
| **DL** | 1.00 | 1.00 | 1.00 | | 1.00 |

## CW-12 - 06/20/2023: Corn, silage

### Sample H69817-02 Lab No. 2355

Sample and source description: Sample H69817-02 Lab No. 2355

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.8 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 11,600.00 | 2,900.00 | 16,800.00 | | 4.11 |
| **DL** | 1.00 | 1.00 | 1.00 | | 1.00 |

## CW-18N - 10/13/2022: Wheat, silage, soft dough

### Sample H51001-03 Lab No. 2106 (Analysis is Field CW-18S, 2023)

Sample and source description: Sample H51001-03 Lab No. 2106 (Analysis is Field CW-18S, 2023)

Sample date: 05/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 50.1 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 6,500.00 | 2,100.00 | 18,800.00 | | 3.93 |
| **DL** | 1.00 | 1.00 | 1.00 | | 1.00 |

## CW-18S - 10/13/2022: Wheat, silage, soft dough

### Sample H51001-03 Lab No. 2106

Sample and source description: Sample H51001-03 Lab No. 2106

Sample date: 05/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 50.1 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 6,500.00 | 2,100.00 | 18,800.00 | | 3.93 |
| **DL** | 1.00 | 1.00 | 1.00 | | 1.00 |

## CW-19N - 10/22/2022: Wheat, silage, soft dough

### Sample H51001-04 Lab No. 2107

Sample and source description: Sample H51001-04 Lab No. 2107

Sample date: 05/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.6 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 9,600.00 | 2,600.00 | 25,900.00 | | 4.16 |
| **DL** | 1.00 | 1.00 | 1.00 | | 1.00 |

## CW-19S - 10/21/2022: Wheat, silage, soft dough

### Sample H51001-05 Lab No. 2108

Sample and source description: Sample H51001-05 Lab No. 2108

Sample date: 05/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.6 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 10,800.00 | 2,700.00 | 26,600.00 | | 4.28 |
| **DL** | 1.00 | 1.00 | 1.00 | | 1.00 |

## CW-19S - 06/14/2023: Corn, silage

### Sample H69817-03 Lab No. 2356

Sample and source description: Sample H69817-03 Lab No. 2356

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.7 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 11,500.00 | 2,100.00 | 13,500.00 | | 3.22 |
| **DL** | 1.00 | 1.00 | 1.00 | | 1.00 |

## CW-20 - 10/21/2022: Wheat, silage, soft dough

### Sample H51001-06 Lab No. 2109

Sample and source description: Sample H51001-06 Lab No. 2109

Sample date: 05/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.0 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 7,300.00 | 2,100.00 | 14,500.00 | | 3.77 |
| **DL** | 1.00 | 1.00 | 1.00 | | 1.00 |

## CW-20 - 06/14/2023: Corn, silage

### Sample H69817-04 Lab No. 2357

Sample and source description: Sample H69817-04 Lab No. 2357

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.2 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 11,000.00 | 2,300.00 | 15,500.00 | | 3.21 |
| **DL** | 1.00 | 1.00 | 1.00 | | 1.00 |

## CW-21 - 10/22/2022: Wheat, silage, soft dough

### Sample H51001-07 Lab No. 2110

Sample and source description: Sample H51001-07 Lab No. 2110

Sample date: 05/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.3 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 13,100.00 | 3,300.00 | 28,400.00 | | 4.44 |
| **DL** | 1.00 | 1.00 | 1.00 | | 1.00 |

## CW-21 - 06/16/2023: Corn, silage

### Sample H69817-05 Lab No. 2358

Sample and source description: Sample H69817-05 Lab No. 2358

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.3 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 7,000.00 | 1,500.00 | 15,400.00 | | 3.70 |
| **DL** | 1.00 | 1.00 | 1.00 | | 1.00 |

## CW-22 - 05/16/2023: Corn, silage

### Sample H69817-05 Lab No. 2358 (Analysis is Field CW-21, 2023)

Sample and source description: Sample H69817-05 Lab No. 2358 (Analysis is Field CW-21, 2023)

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.3 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 7,000.00 | 1,500.00 | 15,400.00 | | 3.70 |
| **DL** | 1.00 | 1.00 | 1.00 | | 1.00 |

## CW-23 - 05/19/2023: Corn, silage

### Sample H69817-05 Lab No. 2358 (Analysis is Field CW-21, 2023)

Sample and source description: Sample H69817-05 Lab No. 2358 (Analysis is Field CW-21, 2023)

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.3 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 7,000.00 | 1,500.00 | 15,400.00 | | 3.70 |
| **DL** | 1.00 | 1.00 | 1.00 | | 1.00 |

## CW-29 - 10/17/2022: Wheat, silage, soft dough

### Sample H51001-08 Lab No. 2111

Sample and source description: Sample H51001-08 Lab No. 2111

Sample date: 05/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 39.2 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 7,200.00 | 2,200.00 | 15,900.00 | | 1.17 |
| **DL** | 1.00 | 1.00 | 1.00 | | 1.00 |

## CW-30 - 11/26/2022: Wheat, silage, soft dough

### Sample H51001-09 Lab No. 2112

Sample and source description: Sample H51001-09 Lab No. 2112

Sample date: 05/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.0 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 10,400.00 | 3,100.00 | 21,800.00 | | 4.08 |
| **DL** | 1.00 | 1.00 | 1.00 | | 1.00 |

## CW-30 - 06/19/2023: Corn, silage

### Sample H69817-06 Lab No. 2359

Sample and source description: Sample H69817-06 Lab No. 2359

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.4 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 9,400.00 | 2,000.00 | 10,700.00 | | 3.46 |
| **DL** | 1.00 | 1.00 | 1.00 | | 1.00 |

## CW-31 - 11/26/2022: Wheat, silage, soft dough

### Sample H51001-10 Lab No. 2113

Sample and source description: Sample H51001-10 Lab No. 2113

Sample date: 05/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.1 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 13,100.00 | 3,100.00 | 25,000.00 | | 3.93 |
| **DL** | 1.00 | 1.00 | 1.00 | | 1.00 |

## CW-31 - 06/20/2023: Corn, silage

### Sample H69817-07 Lab No. 2360

Sample and source description: Sample H69817-07 Lab No. 2360

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.8 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 9,700.00 | 2,700.00 | 12,000.00 | | 3.43 |
| **DL** | 1.00 | 1.00 | 1.00 | | 1.00 |

## CW-9 - 10/27/2022: Wheat, silage, soft dough

### Sample H51001-01 Lab No. 2104

Sample and source description: Sample H51001-01 Lab No. 2104

Sample date: 05/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 51.1 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 9,400.00 | 2,500.00 | 19,700.00 | | 3.88 |
| **DL** | 1.00 | 1.00 | 1.00 | | 1.00 |

## CW-9 - 06/14/2023: Corn, silage

### Sample H69817-01 Lab No. 2354

Sample and source description: Sample H69817-01 Lab No. 2354

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.9 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 9,700.00 | 2,900.00 | 14,000.00 | | 4.21 |
| **DL** | 1.00 | 1.00 | 1.00 | | 1.00 |

## LP-1 - 10/10/2022: Wheat, silage, soft dough

### Sample H51001-11 Lab No. 2114

Sample and source description: Sample H51001-11 Lab No. 2114

Sample date: 05/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.6 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 13,100.00 | 3,300.00 | 29,200.00 | | 4.16 |
| **DL** | 1.00 | 1.00 | 1.00 | | 1.00 |

## LP-1 - 07/09/2023: Corn, silage

### Sample H69817-07 Lab No. 2360 (Analysis is Field CW-31, 2023)

Sample and source description: Sample H69817-07 Lab No. 2360 (Analysis is Field CW-31, 2023)

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.8 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 9,700.00 | 2,700.00 | 12,000.00 | | 3.43 |
| **DL** | 1.00 | 1.00 | 1.00 | | 1.00 |

## LP-2 - 10/11/2022: Wheat, silage, soft dough

### Sample H51001-13 Lab No. 2115

Sample and source description: Sample H51001-13 Lab No. 2115

Sample date: 05/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.2 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 7,000.00 | 2,100.00 | 15,800.00 | | 3.98 |
| **DL** | 1.00 | 1.00 | 1.00 | | 1.00 |

## LP-3 - 10/11/2022: Wheat, silage, soft dough

### Sample H51001-12 Lab No. 2116

Sample and source description: Sample H51001-12 Lab No. 2116

Sample date: 05/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.7 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 9,900.00 | 3,000.00 | 20,400.00 | | 4.01 |
| **DL** | 1.00 | 1.00 | 1.00 | | 1.00 |

LP-4 - 10/12/2022: Wheat, silage, soft dough

Sample H51001-12 Lab No. 2116 (Analysis is Field LP-3, 2023)

Sample and source description: Sample H51001-12 Lab No. 2116 (Analysis is Field LP-3, 2023)

Sample date: 05/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.7 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 9,900.00 | 3,000.00 | 20,400.00 | | 4.01 |
| **DL** | 1.00 | 1.00 | 1.00 | | 1.00 |

LP-5 - 10/12/2022: Wheat, silage, soft dough

Sample H51001-12 Lab No. 2116 (Analysis is Field LP-3, 2023)

Sample and source description: Sample H51001-12 Lab No. 2116 (Analysis is Field LP-3, 2023)

Sample date: 05/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.7 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 9,900.00 | 3,000.00 | 20,400.00 | | 4.01 |
| **DL** | 1.00 | 1.00 | 1.00 | | 1.00 |

## F. SUBSURFACE (TILE) DRAINAGE ANALYSES

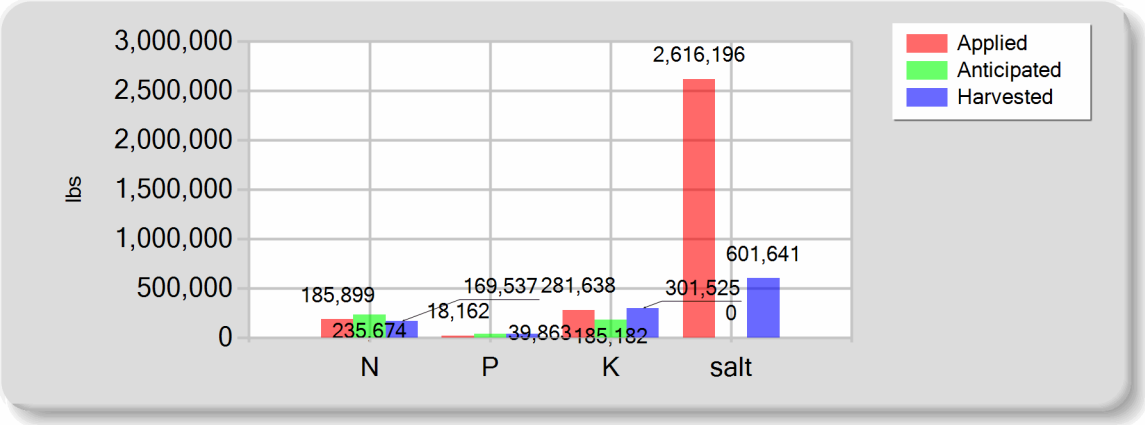
*No subsurface (tile) drainage analyses entered.*

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

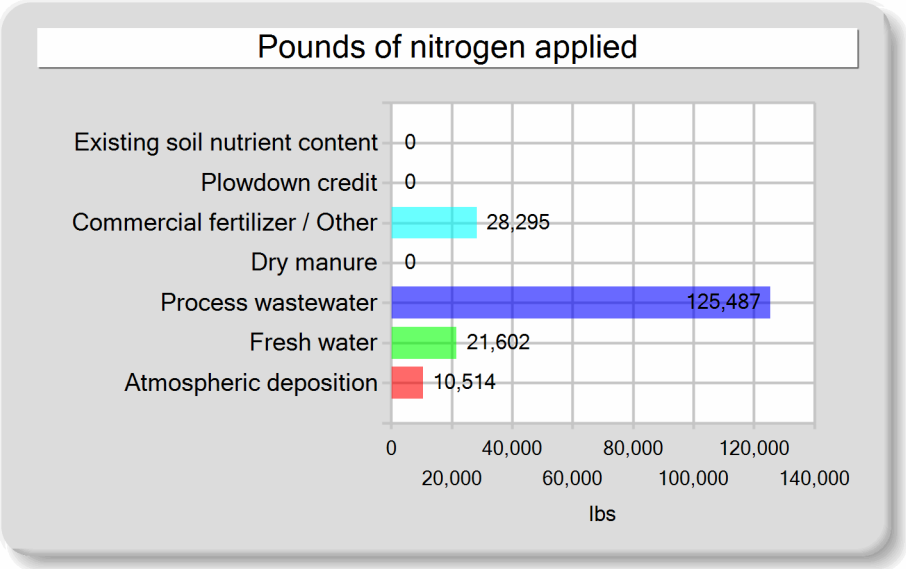
## A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

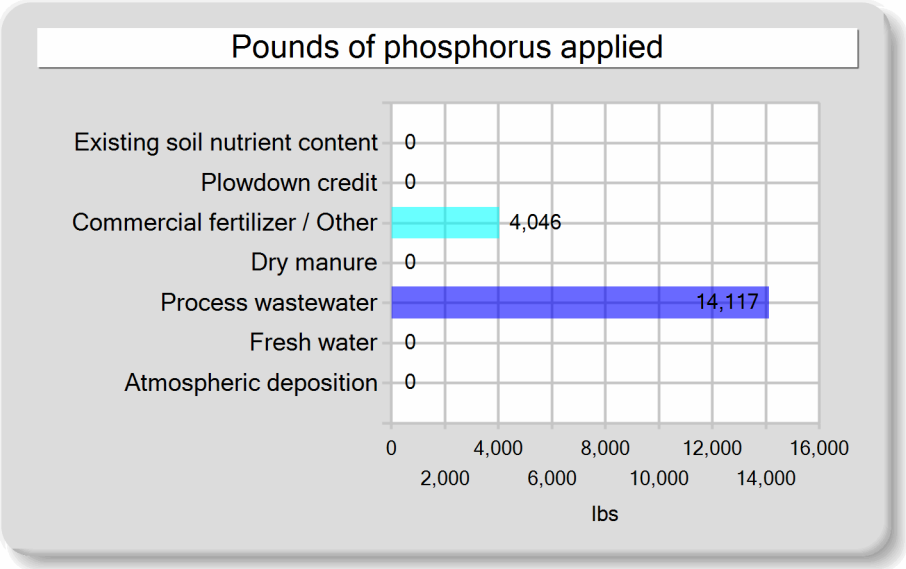
| | Total N (lbs) | Total P (lbs) | Total K (lbs) | Total salt (lbs) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 28,294.80 | 4,045.50 | 2,675.25 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 125,487.43 | 14,116.86 | 278,963.16 | 1,566,579.74 |
| Fresh water | 21,602.47 | 0.00 | 0.00 | 1,049,615.77 |
| Atmospheric deposition | 10,514.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 185,898.70 | 18,162.36 | 281,638.41 | 2,616,195.52 |
| Anticipated crop nutrient removal | 235,674.00 | 39,862.80 | 185,182.20 | 0.00 |
| Actual crop nutrient removal | 169,537.34 | 40,399.00 | 301,525.01 | 601,641.47 |
| Nutrient balance | 16,361.37 | -22,236.64 | -19,886.60 | 2,014,554.05 |
| Applied to removed ratio | 1.10 | 0.45 | 0.93 | 4.35 |

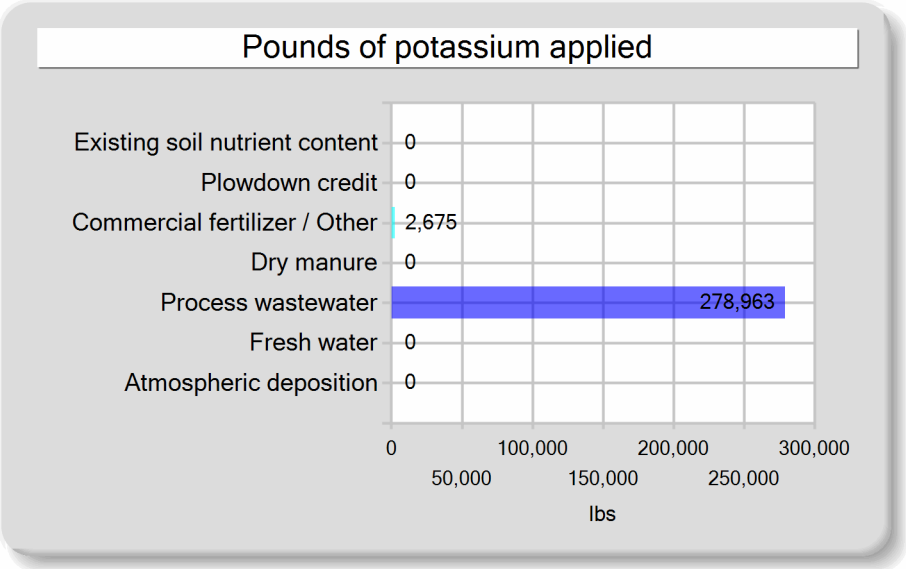
## B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

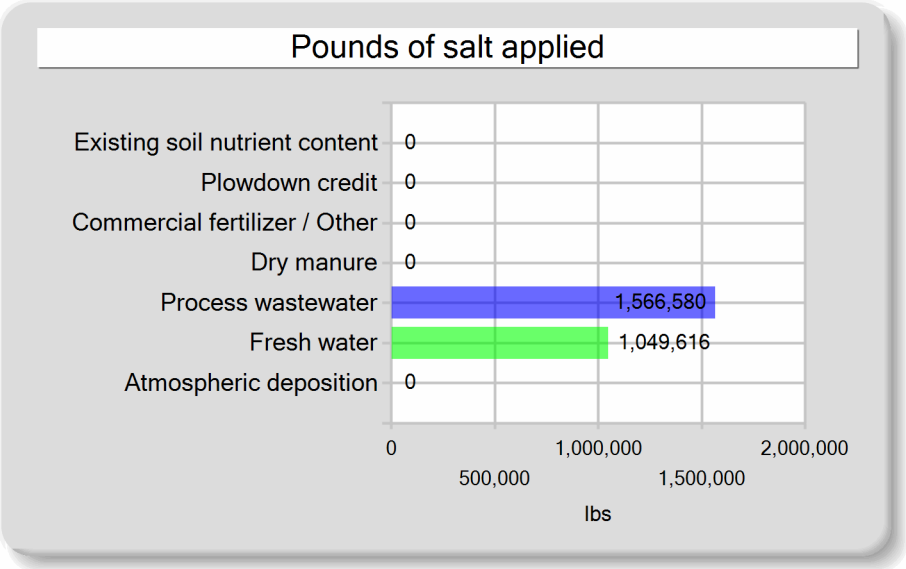


**C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE**



## EXCEPTION REPORTING

### A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

*No manure or process wastewater discharges occurred during the reporting period.*

### B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

*No stormwater discharges occurred during the reporting period.*

### C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

*No land application area to surface water discharges occurred during the reporting period.*

## NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

### A. NUTRIENT MANAGEMENT PLAN STATEMENTS

| | |
|---|---|
| Was the facility's NMP updated in the reporting period? | Yes |
| Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order? | Yes |
| Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order? | Yes |

### B. EXPORT AGREEMENT STATEMENT

| | |
|---|---|
| Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period? | No |

ADDITIONAL NOTES

**A. NOTES**

There are no cows at the dairy. Average milk production (pounds/cow/day) is listed as 1 because the website/portal window could not be “saved” unless average milk production is equal to or greater than 1. Jersey-Holstein Cross is listed as the predominant animal breed because that has been the predominant animal breed historically.

There are no cows at the dairy. Manure excreted by the herd is listed as 1.00 ton(s) per reporting period because the website/portal window could not be “saved” unless there was a value equal to or greater than 1.00. Total nitrogen, Total phosphorus, potassium, and salt from manure are listed a 1.00 pound per reporting period because the website/portal window could not be “saved” unless there was value equal to or greater than 1.

There are no cows at the dairy, so no manure samples were collected for the 2023 reporting year.

The 2023 wheat analyses from field CW-18S was used for the 2023 wheat analyses for field CW-18N. The 2023 wheat analyses from field LP-3 was used for the 2023 wheat analyses for fields LP-4 and LP-5. The 2023 corn analyses from field CW-21 was used for the 2023 corn analyses for fields CW-22 and CW-23. The 2023 corn analyses from field CW-31 was used for the 2023 corn analyses for field LP-1.

The 2022 irrigation well fresh water analyses was used for the 2023 irrigation well fresh water analyses. The 2021 MID (canal) fresh water analyses was used for the 2023 MID fresh water. analyses. The fresh water samples will be collected in the 2024 reporting year.

Fields CW-26 and CW-27 are not included in the 2023 Annual Report due to them being part of the Irrigated Lands Regulatory Program. Fields CW-18N, CW-18S, CW-19N, and CW-29 are not included in the 2023 Annual Report for the 2023 summer crop due to them being part of the Irrigated Lands Regulatory Program. Field CW-22 winter crop (wheat) was disced back into the ground, so no crop was harvested. Fields LP-2, LP-3, LP-4, and LP-5 were fallow for the 2023 summer crop.

The tonnage for Field CW-19S for the 2023 corn harvest event is based on book value.

The commercial fertilizer applied to the corn crop was assumed based on the applications from the 2022 report year.

CERTIFICATION

## A. OWNER AND/OR OPERATOR CERTIFICATION

*I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.*

| | |
|---|---|
| SIGNATURE OF OWNER OF FACILITY | SIGNATURE OF OPERATOR OF FACILITY |
| Michael Gallo | Gallo Cattle Company |
| PRINT OR TYPE NAME | PRINT OR TYPE NAME |
| 6/26/24 | 6/26/24 |
| DATE | DATE |

**Cottonwood Dairy**
10561 W 140 HWY
Atwater, CA 95301

| Groundwater Source ID | Sample Date | Sample ID |
|---|---|---|
| Well 80 | 12/06/2022 | Well 80 |
| Well 82 | 12/06/2022 | Well 82 |
| Well 91.5 | 12/06/2022 | Well 91.5 |
| Well 92 | 12/06/2022 | Well 92 |
| Well 113.5 | 12/06/2022 | Well 113.5 |
| Well 130 | 12/06/2022 | Well 130 |
| Well 131 | 12/06/2022 | Well 131 |
| Well 132 | 12/06/2022 | Well 132 |
| Well 133 | 12/06/2022 | Well 133 |
| Well 134 | 12/06/2022 | Well 134 |
| Well 135 | 12/06/2022 | Well 135 |
| Well 136 | 12/06/2022 | Well 136 |
| Well 137 | 12/06/2022 | Well 137 |
| Well 139 | 12/06/2022 | Well 139 |
| Well 140 | 12/06/2022 | Well 140 |
| Well 144 | 12/06/2022 | Well 144 |
| Well 149 | 12/06/2022 | Well 149 |

**A & L Western Laboratories, Inc.**

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# Ag Suitability Water Analysis

**LOUIE OTT & MORGAN WILLIS**
**GALLO FARMING, LP**
**P.O. BOX 775**
**ATWATER, CA 95301**

**Order Number:** 41114
**Lab Number:** W41114-01
**Submitted Date:** 12/6/2022
**Report Date:** 12/9/2022
**Submitted By:** SERGIO GARCIA
**COC Number:** WELL WATER

## Grower: JOSEPH GALLO FARMS

## Description: WELL 82

mg/L = milligrams/liter = part per million = ppm
meq/L = milliequivalents/liter
dS/m = deciSiemen/meter = mmhos/cm
lbs/ac-ft = pounds/acre-foot

| Analytes | | | | Normal Values | Problem Values |
|---|---|---|---|---|---|
| pH (pH) | | 7.2 | pH units | 6.5 to 8.0 | <6.5 or >8.0 |
| Electrical Conductivity (EC) | | 1.44 | dS/m | 0.5 to 3.0 | <0.5 or >3.0 |
| Boron (B) | | 0.07 | mg/L | <0.5 | >0.5 |
| **Cations** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Calcium (Ca) | 153 | 7.67 | 414 | 30 to 400 | <30 |
| Magnesium (Mg) | 44.8 | 3.73 | 121 | 1 to 60 | [Mg]>[Ca] |
| Sodium (Na) | 91.1 | 3.96 | 246 | <70 | >70 |
| Potassium (K) | 6.9 | 0.18 | 18.7 | | |
| **Anions** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Bicarbonate ($HCO_3$) | 428 | 7.01 | 1150 | <150 | >300 |
| Chloride (Cl) | 86.9 | 2.45 | 235 | <150 | >200 |
| Nitrate - Nitrogen ($NO_3$-N) | 67.2 | 4.80 | 181 | <10 | >10 |
| Sulfate - Sulfur ($SO_4$-S) | 40.6 | 1.27 | 110 | | |
| **Calculated Values** | | | | **Normal Values** | **Problem Values** |
| Total Dissolved Solids (TDS) | | 991 | mg/L | 1 to 1,500 | >1,900 |
| Sodium Absorption Ratio (SAR) | | 1.7 | | <6.0 | >6.0 |
| SAR/EC Ratio (SEC) | | 1.2 | | <5.0 | >10.0 |
| pHc (pHc) | | 6.8 | | <8.4 may add Ca | >8.4 may remove Ca |

| Gypsum Requirements | 100% gypsum equivalent (lbs/ac-ft) |
|---|---|
| Eatons Gypsum Requirement (EGR) | 0 |
| Residual Sodium Carbonate (RSC) | 0 |

Sam Modesitt - Chemist

contact@vtaglab.com

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# Ag Suitability Water Analysis

**LOUIE OTT & MORGAN WILLIS**
**GALLO FARMING, LP**
**P.O. BOX 775**
**ATWATER, CA 95301**

**Order Number:** 41114
**Lab Number:** W41114-02
**Submitted Date:** 12/6/2022
**Report Date:** 12/9/2022
**Submitted By:** SERGIO GARCIA
**COC Number:** WELL WATER

## Grower: JOSEPH GALLO FARMS

## Description: WELL 80

mg/L = milligrams/liter = part per million = ppm
meq/L = milliequivalents/liter
dS/m = deciSiemen/meter = mmhos/cm
lbs/ac-ft = pounds/acre-foot

| Analytes | | | | Normal Values | Problem Values |
|---|---|---|---|---|---|
| pH (pH) | | 7.4 | pH units | 6.5 to 8.0 | <6.5 or >8.0 |
| Electrical Conductivity (EC) | | 0.90 | dS/m | 0.5 to 3.0 | <0.5 or >3.0 |
| Boron (B) | | 0.03 | mg/L | <0.5 | >0.5 |
| **Cations** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Calcium (Ca) | 97.8 | 4.89 | 264 | 30 to 400 | <30 |
| Magnesium (Mg) | 26.9 | 2.24 | 72.6 | 1 to 60 | [Mg]>[Ca] |
| Sodium (Na) | 40.7 | 1.77 | 110 | <70 | >70 |
| Potassium (K) | 4.8 | 0.12 | 13.1 | | |
| **Anions** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Bicarbonate ($HCO_3$) | 254 | 4.17 | 687 | <150 | >300 |
| Chloride (Cl) | 39.3 | 1.11 | 106 | <150 | >200 |
| Nitrate - Nitrogen ($NO_3$-N) | 31.6 | 2.26 | 85.4 | <10 | >10 |
| Sulfate - Sulfur ($SO_4$-S) | 45.4 | 1.42 | 123 | | |
| **Calculated Values** | | | | **Normal Values** | **Problem Values** |
| Total Dissolved Solids (TDS) | | 616 | mg/L | 1 to 1,500 | >1,900 |
| Sodium Absorption Ratio (SAR) | | 0.9 | | <6.0 | >6.0 |
| SAR/EC Ratio (SEC) | | 1.1 | | <5.0 | >10.0 |
| pHc (pHc) | | 7.2 | | <8.4 may add Ca | >8.4 may remove Ca |

| Gypsum Requirements | 100% gypsum equivalent (lbs/ac-ft) |
|---|---|
| Eatons Gypsum Requirement (EGR) | 0 |
| Residual Sodium Carbonate (RSC) | 0 |

Sam Modesitt - Chemist

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# Ag Suitability Water Analysis

**LOUIE OTT & MORGAN WILLIS**
**GALLO FARMING, LP**
**P.O. BOX 775**
**ATWATER, CA 95301**

**Order Number:** 41113
**Lab Number:** W41113-01
**Submitted Date:** 12/6/2022
**Report Date:** 12/9/2022
**Submitted By:** SERGIO GARCIA
**COC Number:** WELL WATER

## Grower: JOSEPH GALLO FARMS

## Description: WELL 140

mg/L = milligrams/liter = part per million = ppm
meq/L = milliequivalents/liter
dS/m = deciSiemen/meter = mmhos/cm
lbs/ac-ft = pounds/acre-foot

| Analytes | | | | Normal Values | Problem Values |
|---|---|---|---|---|---|
| pH (pH) | | 7.6 | pH units | 6.5 to 8.0 | <6.5 or >8.0 |
| Electrical Conductivity (EC) | | 1.05 | dS/m | 0.5 to 3.0 | <0.5 or >3.0 |
| Boron (B) | | 0.04 | mg/L | <0.5 | >0.5 |
| **Cations** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Calcium (Ca) | 111 | 5.55 | 300 | 30 to 400 | <30 |
| Magnesium (Mg) | 31.0 | 2.58 | 83.6 | 1 to 60 | [Mg]>[Ca] |
| Sodium (Na) | 78.4 | 3.41 | 212 | <70 | >70 |
| Potassium (K) | 4.2 | 0.11 | 11.4 | | |
| **Anions** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Bicarbonate ($HCO_3$) | 448 | 7.34 | 1210 | <150 | >300 |
| Chloride (Cl) | 65.6 | 1.85 | 177 | <150 | >200 |
| Nitrate - Nitrogen ($NO_3$-N) | 13.9 | 1.00 | 37.6 | <10 | >10 |
| Sulfate - Sulfur ($SO_4$-S) | 49.0 | 1.53 | 132 | | |
| **Calculated Values** | | | | **Normal Values** | **Problem Values** |
| Total Dissolved Solids (TDS) | | 722 | mg/L | 1 to 1,500 | >1,900 |
| Sodium Absorption Ratio (SAR) | | 1.7 | | <6.0 | >6.0 |
| SAR/EC Ratio (SEC) | | 1.6 | | <5.0 | >10.0 |
| pHc (pHc) | | 6.9 | | <8.4 may add Ca | >8.4 may remove Ca |

| Gypsum Requirements | 100% gypsum equivalent (lbs/ac-ft) |
|---|---|
| Eatons Gypsum Requirement (EGR) | 0 |
| Residual Sodium Carbonate (RSC) | 0 |

Sam Modesitt - Chemist

contact@vtaglab.com

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# Ag Suitability Water Analysis

**LOUIE OTT & MORGAN WILLIS**
**GALLO FARMING, LP**
**P.O. BOX 775**
**ATWATER, CA 95301**

| | |
|---|---|
| **Order Number:** | **41113** |
| **Lab Number:** | **W41113-02** |
| **Submitted Date:** | **12/6/2022** |
| **Report Date:** | **12/9/2022** |
| **Submitted By:** | **SERGIO GARCIA** |
| **COC Number:** | **WELL WATER** |

## Grower: JOSEPH GALLO FARMS

## Description: WELL 135

mg/L = milligrams/liter = part per million = ppm
meq/L = milliequivalents/liter
dS/m = deciSiemen/meter = mmhos/cm
lbs/ac-ft = pounds/acre-foot

| Analytes | | | | Normal Values | Problem Values |
|---|---|---|---|---|---|
| pH (pH) | | 7.8 | pH units | 6.5 to 8.0 | <6.5 or >8.0 |
| Electrical Conductivity (EC) | | 0.85 | dS/m | 0.5 to 3.0 | <0.5 or >3.0 |
| Boron (B) | | 0.03 | mg/L | <0.5 | >0.5 |
| **Cations** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Calcium (Ca) | 91.2 | 4.56 | 246 | 30 to 400 | <30 |
| Magnesium (Mg) | 18.0 | 1.50 | 48.6 | 1 to 60 | [Mg]>[Ca] |
| Sodium (Na) | 71.3 | 3.10 | 193 | <70 | >70 |
| Potassium (K) | 3.3 | 0.08 | 8.8 | | |
| **Anions** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Bicarbonate ($HCO_3$) | 371 | 6.08 | 1000 | <150 | >300 |
| Chloride (Cl) | 41.8 | 1.18 | 113 | <150 | >200 |
| Nitrate - Nitrogen ($NO_3$-N) | 1.1 | 0.08 | 2.9 | <10 | >10 |
| Sulfate - Sulfur ($SO_4$-S) | 60.5 | 1.89 | 163 | | |
| **Calculated Values** | | | | **Normal Values** | **Problem Values** |
| Total Dissolved Solids (TDS) | | 585 | mg/L | 1 to 1,500 | >1,900 |
| Sodium Absorption Ratio (SAR) | | 1.8 | | <6.0 | >6.0 |
| SAR/EC Ratio (SEC) | | 2.1 | | <5.0 | >10.0 |
| pHc (pHc) | | 7.2 | | <8.4 may add Ca | >8.4 may remove Ca |

| Gypsum Requirements | 100% gypsum equivalent (lbs/ac-ft) |
|---|---|
| Eatons Gypsum Requirement (EGR) | 0 |
| Residual Sodium Carbonate (RSC) | 5 |

Sam Modesitt - Chemist

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# Ag Suitability Water Analysis

**LOUIE OTT & MORGAN WILLIS**
**GALLO FARMING, LP**
**P.O. BOX 775**
**ATWATER, CA 95301**

**Order Number:** 41113
**Lab Number:** W41113-03
**Submitted Date:** 12/6/2022
**Report Date:** 12/9/2022
**Submitted By:** SERGIO GARCIA
**COC Number:** WELL WATER

## Grower: JOSEPH GALLO FARMS

## Description: WELL 134

mg/L = milligrams/liter = part per million = ppm
meq/L = milliequivalents/liter
dS/m = deciSiemen/meter = mmhos/cm
lbs/ac-ft = pounds/acre-foot

| Analytes | | | | Normal Values | Problem Values |
|---|---|---|---|---|---|
| pH (pH) | | 7.5 | pH units | 6.5 to 8.0 | <6.5 or >8.0 |
| Electrical Conductivity (EC) | | 1.21 | dS/m | 0.5 to 3.0 | <0.5 or >3.0 |
| Boron (B) | | 0.04 | mg/L | <0.5 | >0.5 |
| **Cations** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Calcium (Ca) | 118 | 5.90 | 319 | 30 to 400 | <30 |
| Magnesium (Mg) | 38.0 | 3.17 | 103 | 1 to 60 | [Mg]>[Ca] |
| Sodium (Na) | 100 | 4.35 | 270 | <70 | >70 |
| Potassium (K) | 3.9 | 0.10 | 10.5 | | |
| **Anions** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Bicarbonate ($HCO_3$) | 437 | 7.17 | 1180 | <150 | >300 |
| Chloride (Cl) | 52.1 | 1.47 | 141 | <150 | >200 |
| Nitrate - Nitrogen ($NO_3$-N) | 11.0 | 0.79 | 29.7 | <10 | >10 |
| Sulfate - Sulfur ($SO_4$-S) | 130 | 4.05 | 350 | | |
| **Calculated Values** | | | | **Normal Values** | **Problem Values** |
| Total Dissolved Solids (TDS) | | 832 | mg/L | 1 to 1,500 | >1,900 |
| Sodium Absorption Ratio (SAR) | | 2.0 | | <6.0 | >6.0 |
| SAR/EC Ratio (SEC) | | 1.7 | | <5.0 | >10.0 |
| pHc (pHc) | | 6.9 | | <8.4 may add Ca | >8.4 may remove Ca |

| Gypsum Requirements | 100% gypsum equivalent (lbs/ac-ft) |
|---|---|
| Eatons Gypsum Requirement (EGR) | 0 |
| Residual Sodium Carbonate (RSC) | 0 |

Sam Modesitt - Chemist

contact@vtaglab.com

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# Ag Suitability Water Analysis

**LOUIE OTT & MORGAN WILLIS**
**GALLO FARMING, LP**
**P.O. BOX 775**
**ATWATER, CA 95301**

**Order Number:** 41113
**Lab Number:** W41113-04
**Submitted Date:** 12/6/2022
**Report Date:** 12/9/2022
**Submitted By:** SERGIO GARCIA
**COC Number:** WELL WATER

## Grower: JOSEPH GALLO FARMS

## Description: WELL 133

mg/L = milligrams/liter = part per million = ppm
meq/L = milliequivalents/liter
dS/m = deciSiemen/meter = mmhos/cm
lbs/ac-ft = pounds/acre-foot

| Analytes | | | | Normal Values | Problem Values |
|---|---|---|---|---|---|
| pH (pH) | | 7.4 | pH units | 6.5 to 8.0 | <6.5 or >8.0 |
| Electrical Conductivity (EC) | | 1.13 | dS/m | 0.5 to 3.0 | <0.5 or >3.0 |
| Boron (B) | | 0.03 | mg/L | <0.5 | >0.5 |
| **Cations** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Calcium (Ca) | 118 | 5.89 | 318 | 30 to 400 | <30 |
| Magnesium (Mg) | 39.1 | 3.26 | 106 | 1 to 60 | [Mg]>[Ca] |
| Sodium (Na) | 74.8 | 3.25 | 202 | <70 | >70 |
| Potassium (K) | 3.8 | 0.10 | 10.3 | | |
| **Anions** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Bicarbonate ($HCO_3$) | 464 | 7.61 | 1250 | <150 | >300 |
| Chloride (Cl) | 40.1 | 1.13 | 108 | <150 | >200 |
| Nitrate - Nitrogen ($NO_3$-N) | 8.7 | 0.62 | 23.5 | <10 | >10 |
| Sulfate - Sulfur ($SO_4$-S) | 98.6 | 3.08 | 266 | | |
| **Calculated Values** | | | | **Normal Values** | **Problem Values** |
| Total Dissolved Solids (TDS) | | 777 | mg/L | 1 to 1,500 | >1,900 |
| Sodium Absorption Ratio (SAR) | | 1.5 | | <6.0 | >6.0 |
| SAR/EC Ratio (SEC) | | 1.4 | | <5.0 | >10.0 |
| pHc (pHc) | | 6.9 | | <8.4 may add Ca | >8.4 may remove Ca |

| Gypsum Requirements | 100% gypsum equivalent (lbs/ac-ft) |
|---|---|
| Eatons Gypsum Requirement (EGR) | 0 |
| Residual Sodium Carbonate (RSC) | 0 |

Sam Modesitt - Chemist

contact@vtaglab.com

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# Ag Suitability Water Analysis

**LOUIE OTT & MORGAN WILLIS**
**GALLO FARMING, LP**
**P.O. BOX 775**
**ATWATER, CA 95301**

**Order Number:** 41113
**Lab Number:** W41113-05
**Submitted Date:** 12/6/2022
**Report Date:** 12/9/2022
**Submitted By:** SERGIO GARCIA
**COC Number:** WELL WATER

## Grower: JOSEPH GALLO FARMS

## Description: WELL 132

mg/L = milligrams/liter = part per million = ppm
meq/L = milliequivalents/liter
dS/m = deciSiemen/meter = mmhos/cm
lbs/ac-ft = pounds/acre-foot

| Analytes | | | | Normal Values | Problem Values |
|---|---|---|---|---|---|
| pH (pH) | | 7.4 | pH units | 6.5 to 8.0 | <6.5 or >8.0 |
| Electrical Conductivity (EC) | | 1.21 | dS/m | 0.5 to 3.0 | <0.5 or >3.0 |
| Boron (B) | | 0.04 | mg/L | <0.5 | >0.5 |
| **Cations** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Calcium (Ca) | 119 | 5.94 | 321 | 30 to 400 | <30 |
| Magnesium (Mg) | 35.9 | 2.99 | 96.9 | 1 to 60 | [Mg]>[Ca] |
| Sodium (Na) | 97.3 | 4.23 | 263 | <70 | >70 |
| Potassium (K) | 4.5 | 0.11 | 12.0 | | |
| **Anions** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Bicarbonate ($HCO_3$) | 495 | 8.12 | 1340 | <150 | >300 |
| Chloride (Cl) | 64.5 | 1.82 | 174 | <150 | >200 |
| Nitrate - Nitrogen ($NO_3$-N) | 17.9 | 1.28 | 48.4 | <10 | >10 |
| Sulfate - Sulfur ($SO_4$-S) | 66.9 | 2.09 | 181 | | |
| **Calculated Values** | | | | **Normal Values** | **Problem Values** |
| Total Dissolved Solids (TDS) | | 832 | mg/L | 1 to 1,500 | >1,900 |
| Sodium Absorption Ratio (SAR) | | 2.0 | | <6.0 | >6.0 |
| SAR/EC Ratio (SEC) | | 1.7 | | <5.0 | >10.0 |
| pHc (pHc) | | 6.8 | | <8.4 may add Ca | >8.4 may remove Ca |

| Gypsum Requirements | 100% gypsum equivalent (lbs/ac-ft) |
|---|---|
| Eatons Gypsum Requirement (EGR) | 0 |
| Residual Sodium Carbonate (RSC) | 0 |

Sam Modesitt - Chemist

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# Ag Suitability Water Analysis

**LOUIE OTT & MORGAN WILLIS**
**GALLO FARMING, LP**
**P.O. BOX 775**
**ATWATER, CA 95301**

| | |
|---|---|
| **Order Number:** | **41113** |
| **Lab Number:** | **W41113-06** |
| **Submitted Date:** | **12/6/2022** |
| **Report Date:** | **12/9/2022** |
| **Submitted By:** | **SERGIO GARCIA** |
| **COC Number:** | **WELL WATER** |

## Grower: JOSEPH GALLO FARMS

## Description: WELL 131

mg/L = milligrams/liter = part per million = ppm
meq/L = milliequivalents/liter
dS/m = deciSiemen/meter = mmhos/cm
lbs/ac-ft = pounds/acre-foot

| Analytes | | | | Normal Values | Problem Values |
|---|---|---|---|---|---|
| pH (pH) | | 7.1 | pH units | 6.5 to 8.0 | <6.5 or >8.0 |
| Electrical Conductivity (EC) | | 1.27 | dS/m | 0.5 to 3.0 | <0.5 or >3.0 |
| Boron (B) | | 0.03 | mg/L | <0.5 | >0.5 |
| **Cations** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Calcium (Ca) | 138 | 6.91 | 373 | 30 to 400 | <30 |
| Magnesium (Mg) | 50.0 | 4.17 | 135 | 1 to 60 | [Mg]>[Ca] |
| Sodium (Na) | 80.7 | 3.51 | 218 | <70 | >70 |
| Potassium (K) | 3.8 | 0.10 | 10.2 | | |
| **Anions** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Bicarbonate ($HCO_3$) | 459 | 7.52 | 1240 | <150 | >300 |
| Chloride (Cl) | 34.0 | 0.96 | 91.8 | <150 | >200 |
| Nitrate - Nitrogen ($NO_3$-N) | 7.2 | 0.51 | 19.4 | <10 | >10 |
| Sulfate - Sulfur ($SO_4$-S) | 177 | 5.52 | 477 | | |
| **Calculated Values** | | | | **Normal Values** | **Problem Values** |
| Total Dissolved Solids (TDS) | | 874 | mg/L | 1 to 1,500 | >1,900 |
| Sodium Absorption Ratio (SAR) | | 1.5 | | <6.0 | >6.0 |
| SAR/EC Ratio (SEC) | | 1.2 | | <5.0 | >10.0 |
| pHc (pHc) | | 6.8 | | <8.4 may add Ca | >8.4 may remove Ca |

| Gypsum Requirements | 100% gypsum equivalent (lbs/ac-ft) |
|---|---|
| Eatons Gypsum Requirement (EGR) | 0 |
| Residual Sodium Carbonate (RSC) | 0 |

Sam Modesitt - Chemist

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# Ag Suitability Water Analysis

**LOUIE OTT & MORGAN WILLIS**
**GALLO FARMING, LP**
**P.O. BOX 775**
**ATWATER, CA 95301**

| | |
|---|---|
| **Order Number:** | **41113** |
| **Lab Number:** | **W41113-07** |
| **Submitted Date:** | **12/6/2022** |
| **Report Date:** | **12/9/2022** |
| **Submitted By:** | **SERGIO GARCIA** |
| **COC Number:** | **WELL WATER** |

## Grower: JOSEPH GALLO FARMS

## Description: WELL 130

mg/L = milligrams/liter = part per million = ppm
meq/L = milliequivalents/liter
dS/m = deciSiemen/meter = mmhos/cm
lbs/ac-ft = pounds/acre-foot

| Analytes | | | | Normal Values | Problem Values |
|---|---|---|---|---|---|
| pH (pH) | | 7.1 | pH units | 6.5 to 8.0 | <6.5 or >8.0 |
| Electrical Conductivity (EC) | | 1.59 | dS/m | 0.5 to 3.0 | <0.5 or >3.0 |
| Boron (B) | | 0.09 | mg/L | <0.5 | >0.5 |
| **Cations** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Calcium (Ca) | 142 | 7.10 | 383 | 30 to 400 | <30 |
| Magnesium (Mg) | 46.0 | 3.83 | 124 | 1 to 60 | [Mg]>[Ca] |
| Sodium (Na) | 134 | 5.82 | 361 | <70 | >70 |
| Potassium (K) | 5.3 | 0.14 | 14.4 | | |
| **Anions** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Bicarbonate ($HCO_3$) | 475 | 7.79 | 1280 | <150 | >300 |
| Chloride (Cl) | 122 | 3.44 | 329 | <150 | >200 |
| Nitrate - Nitrogen ($NO_3$-N) | 54.3 | 3.88 | 147 | <10 | >10 |
| Sulfate - Sulfur ($SO_4$-S) | 58.6 | 1.83 | 158 | | |
| **Calculated Values** | | | | **Normal Values** | **Problem Values** |
| Total Dissolved Solids (TDS) | | 1090 | mg/L | 1 to 1,500 | >1,900 |
| Sodium Absorption Ratio (SAR) | | 2.5 | | <6.0 | >6.0 |
| SAR/EC Ratio (SEC) | | 1.6 | | <5.0 | >10.0 |
| pHc (pHc) | | 6.8 | | <8.4 may add Ca | >8.4 may remove Ca |

| Gypsum Requirements | 100% gypsum equivalent (lbs/ac-ft) |
|---|---|
| Eatons Gypsum Requirement (EGR) | 0 |
| Residual Sodium Carbonate (RSC) | 0 |

Sam Modesitt - Chemist

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# Ag Suitability Water Analysis

**LOUIE OTT & MORGAN WILLIS**
**GALLO FARMING, LP**
**P.O. BOX 775**
**ATWATER, CA 95301**

| | |
|---|---|
| **Order Number:** | **41113** |
| **Lab Number:** | **W41113-08** |
| **Submitted Date:** | **12/6/2022** |
| **Report Date:** | **12/9/2022** |
| **Submitted By:** | **SERGIO GARCIA** |
| **COC Number:** | **WELL WATER** |

## Grower: JOSEPH GALLO FARMS

## Description: WELL 136

mg/L = milligrams/liter = part per million = ppm
meq/L = milliequivalents/liter
dS/m = deciSiemen/meter = mmhos/cm
lbs/ac-ft = pounds/acre-foot

| Analytes | | | | Normal Values | Problem Values |
|---|---|---|---|---|---|
| pH (pH) | | 7.5 | pH units | 6.5 to 8.0 | <6.5 or >8.0 |
| Electrical Conductivity (EC) | | 0.63 | dS/m | 0.5 to 3.0 | <0.5 or >3.0 |
| Boron (B) | | 0.04 | mg/L | <0.5 | >0.5 |
| **Cations** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Calcium (Ca) | 46.8 | 2.34 | 126 | 30 to 400 | <30 |
| Magnesium (Mg) | 9.5 | 0.79 | 25.5 | 1 to 60 | [Mg]>[Ca] |
| Sodium (Na) | 74.5 | 3.24 | 201 | <70 | >70 |
| Potassium (K) | 1.9 | 0.05 | 5.2 | | |
| **Anions** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Bicarbonate ($HCO_3$) | 258 | 4.23 | 697 | <150 | >300 |
| Chloride (Cl) | 29.9 | 0.84 | 80.8 | <150 | >200 |
| Nitrate - Nitrogen ($NO_3$-N) | 6.5 | 0.46 | 17.4 | <10 | >10 |
| Sulfate - Sulfur ($SO_4$-S) | 28.8 | 0.90 | 77.8 | | |
| **Calculated Values** | | | | **Normal Values** | **Problem Values** |
| Total Dissolved Solids (TDS) | | 431 | mg/L | 1 to 1,500 | >1,900 |
| Sodium Absorption Ratio (SAR) | | 2.6 | | <6.0 | >6.0 |
| SAR/EC Ratio (SEC) | | 4.1 | | <5.0 | >10.0 |
| pHc (pHc) | | 7.5 | | <8.4 may add Ca | >8.4 may remove Ca |

| Gypsum Requirements | 100% gypsum equivalent (lbs/ac-ft) |
|---|---|
| Eatons Gypsum Requirement (EGR) | 357 |
| Residual Sodium Carbonate (RSC) | 258 |

Sam Modesitt - Chemist

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# Ag Suitability Water Analysis

**LOUIE OTT & MORGAN WILLIS**
**GALLO FARMING, LP**
**P.O. BOX 775**
**ATWATER, CA 95301**

**Order Number:** 41113
**Lab Number:** W41113-09
**Submitted Date:** 12/6/2022
**Report Date:** 12/9/2022
**Submitted By:** SERGIO GARCIA
**COC Number:** WELL WATER

## Grower: JOSEPH GALLO FARMS

## Description: WELL 137

mg/L = milligrams/liter = part per million = ppm
meq/L = milliequivalents/liter
dS/m = deciSiemen/meter = mmhos/cm
lbs/ac-ft = pounds/acre-foot

| Analytes | | | | Normal Values | Problem Values |
|---|---|---|---|---|---|
| pH (pH) | | 7.6 | pH units | 6.5 to 8.0 | <6.5 or >8.0 |
| Electrical Conductivity (EC) | | 0.59 | dS/m | 0.5 to 3.0 | <0.5 or >3.0 |
| Boron (B) | | 0.05 | mg/L | <0.5 | >0.5 |
| **Cations** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Calcium (Ca) | 46.2 | 2.31 | 125 | 30 to 400 | <30 |
| Magnesium (Mg) | 11.0 | 0.92 | 29.7 | 1 to 60 | [Mg]>[Ca] |
| Sodium (Na) | 69.9 | 3.04 | 189 | <70 | >70 |
| Potassium (K) | 2.3 | 0.06 | 6.3 | | |
| **Anions** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Bicarbonate ($HCO_3$) | 282 | 4.62 | 761 | <150 | >300 |
| Chloride (Cl) | 20.4 | 0.58 | 55.0 | <150 | >200 |
| Nitrate - Nitrogen ($NO_3$-N) | 3.8 | 0.27 | 10.1 | <10 | >10 |
| Sulfate - Sulfur ($SO_4$-S) | 23.1 | 0.72 | 62.5 | | |
| **Calculated Values** | | | | **Normal Values** | **Problem Values** |
| Total Dissolved Solids (TDS) | | 404 | mg/L | 1 to 1,500 | >1,900 |
| Sodium Absorption Ratio (SAR) | | 2.4 | | <6.0 | >6.0 |
| SAR/EC Ratio (SEC) | | 4.1 | | <5.0 | >10.0 |
| pHc (pHc) | | 7.4 | | <8.4 may add Ca | >8.4 may remove Ca |

| Gypsum Requirements | 100% gypsum equivalent (lbs/ac-ft) |
|---|---|
| Eatons Gypsum Requirement (EGR) | 378 |
| Residual Sodium Carbonate (RSC) | 326 |

Sam Modesitt - Chemist

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# Ag Suitability Water Analysis

**LOUIE OTT & MORGAN WILLIS**
**GALLO FARMING, LP**
**P.O. BOX 775**
**ATWATER, CA 95301**

**Order Number:** 41113
**Lab Number:** W41113-10
**Submitted Date:** 12/6/2022
**Report Date:** 12/9/2022
**Submitted By:** SERGIO GARCIA
**COC Number:** WELL WATER

## Grower: JOSEPH GALLO FARMS

## Description: WELL 139

mg/L = milligrams/liter = part per million = ppm
meq/L = milliequivalents/liter
dS/m = deciSiemen/meter = mmhos/cm
lbs/ac-ft = pounds/acre-foot

| Analytes | | | | Normal Values | Problem Values |
|---|---|---|---|---|---|
| pH (pH) | | 8.0 | pH units | 6.5 to 8.0 | <6.5 or >8.0 |
| Electrical Conductivity (EC) | | 0.32 | dS/m | 0.5 to 3.0 | <0.5 or >3.0 |
| Boron (B) | | 0.05 | mg/L | <0.5 | >0.5 |
| **Cations** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Calcium (Ca) | 7.3 | 0.37 | 19.8 | 30 to 400 | <30 |
| Magnesium (Mg) | 1.2 | 0.10 | 3.3 | 1 to 60 | [Mg]>[Ca] |
| Sodium (Na) | 59.6 | 2.59 | 161 | <70 | >70 |
| Potassium (K) | 1.9 | 0.05 | 5.1 | | |
| **Anions** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Bicarbonate ($HCO_3$) | 137 | 2.25 | 371 | <150 | >300 |
| Chloride (Cl) | 19.3 | 0.54 | 52.1 | <150 | >200 |
| Nitrate - Nitrogen ($NO_3$-N) | 0.7 | 0.05 | 2.0 | <10 | >10 |
| Sulfate - Sulfur ($SO_4$-S) | 12.4 | 0.39 | 33.4 | | |
| **Calculated Values** | | | | **Normal Values** | **Problem Values** |
| Total Dissolved Solids (TDS) | | 218 | mg/L | 1 to 1,500 | >1,900 |
| Sodium Absorption Ratio (SAR) | | 5.3 | | <6.0 | >6.0 |
| SAR/EC Ratio (SEC) | | 16.8 | | <5.0 | >10.0 |
| pHc (pHc) | | 8.6 | | <8.4 may add Ca | >8.4 may remove Ca |

| Gypsum Requirements | 100% gypsum equivalent (lbs/ac-ft) |
|---|---|
| Eatons Gypsum Requirement (EGR) | 589 |
| Residual Sodium Carbonate (RSC) | 417 |

Sam Modesitt - Chemist

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# Ag Suitability Water Analysis

**LOUIE OTT & MORGAN WILLIS**
**GALLO FARMING, LP**
**P.O. BOX 775**
**ATWATER, CA 95301**

**Order Number:** 41113
**Lab Number:** W41113-11
**Submitted Date:** 12/6/2022
**Report Date:** 12/9/2022
**Submitted By:** SERGIO GARCIA
**COC Number:** WELL WATER

## Grower: JOSEPH GALLO FARMS

## Description: WELL 149

mg/L = milligrams/liter = part per million = ppm
meq/L = milliequivalents/liter
dS/m = deciSiemen/meter = mmhos/cm
lbs/ac-ft = pounds/acre-foot

| Analytes | | | | Normal Values | Problem Values |
|---|---|---|---|---|---|
| pH (pH) | | 7.5 | pH units | 6.5 to 8.0 | <6.5 or >8.0 |
| Electrical Conductivity (EC) | | 0.72 | dS/m | 0.5 to 3.0 | <0.5 or >3.0 |
| Boron (B) | | 0.08 | mg/L | <0.5 | >0.5 |
| **Cations** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Calcium (Ca) | 58.0 | 2.90 | 157 | 30 to 400 | <30 |
| Magnesium (Mg) | 16.3 | 1.36 | 44.1 | 1 to 60 | [Mg]>[Ca] |
| Sodium (Na) | 82.6 | 3.59 | 223 | <70 | >70 |
| Potassium (K) | 4.2 | 0.11 | 11.3 | | |
| **Anions** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Bicarbonate ($HCO_3$) | 337 | 5.52 | 909 | <150 | >300 |
| Chloride (Cl) | 25.8 | 0.73 | 69.8 | <150 | >200 |
| Nitrate - Nitrogen ($NO_3$-N) | 10.0 | 0.71 | 26.9 | <10 | >10 |
| Sulfate - Sulfur ($SO_4$-S) | 28.1 | 0.88 | 75.9 | | |
| **Calculated Values** | | | | **Normal Values** | **Problem Values** |
| Total Dissolved Solids (TDS) | | 496 | mg/L | 1 to 1,500 | >1,900 |
| Sodium Absorption Ratio (SAR) | | 2.5 | | <6.0 | >6.0 |
| SAR/EC Ratio (SEC) | | 3.4 | | <5.0 | >10.0 |
| pHc (pHc) | | 7.3 | | <8.4 may add Ca | >8.4 may remove Ca |

| Gypsum Requirements | 100% gypsum equivalent (lbs/ac-ft) |
|---|---|
| Eatons Gypsum Requirement (EGR) | 339 |
| Residual Sodium Carbonate (RSC) | 295 |

Sam Modesitt - Chemist

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# Ag Suitability Water Analysis

**LOUIE OTT & MORGAN WILLIS**
**GALLO FARMING, LP**
**P.O. BOX 775**
**ATWATER, CA 95301**

| | |
|---|---|
| **Order Number:** | **41113** |
| **Lab Number:** | **W41113-12** |
| **Submitted Date:** | **12/6/2022** |
| **Report Date:** | **12/9/2022** |
| **Submitted By:** | **SERGIO GARCIA** |
| **COC Number:** | **WELL WATER** |

## Grower: JOSEPH GALLO FARMS

## Description: WELL 144

mg/L = milligrams/liter = part per million = ppm
meq/L = milliequivalents/liter
dS/m = deciSiemen/meter = mmhos/cm
lbs/ac-ft = pounds/acre-foot

| Analytes | | | | Normal Values | Problem Values |
|---|---|---|---|---|---|
| pH (pH) | | 7.5 | pH units | 6.5 to 8.0 | <6.5 or >8.0 |
| Electrical Conductivity (EC) | | 1.07 | dS/m | 0.5 to 3.0 | <0.5 or >3.0 |
| Boron (B) | | 0.05 | mg/L | <0.5 | >0.5 |
| **Cations** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Calcium (Ca) | 108 | 5.40 | 292 | 30 to 400 | <30 |
| Magnesium (Mg) | 32.6 | 2.72 | 88.1 | 1 to 60 | [Mg]>[Ca] |
| Sodium (Na) | 86.0 | 3.74 | 232 | <70 | >70 |
| Potassium (K) | 4.1 | 0.11 | 11.2 | | |
| **Anions** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Bicarbonate ($HCO_3$) | 461 | 7.56 | 1250 | <150 | >300 |
| Chloride (Cl) | 64.5 | 1.82 | 174 | <150 | >200 |
| Nitrate - Nitrogen ($NO_3$-N) | 8.1 | 0.58 | 22.0 | <10 | >10 |
| Sulfate - Sulfur ($SO_4$-S) | 58.6 | 1.83 | 158 | | |
| **Calculated Values** | | | | **Normal Values** | **Problem Values** |
| Total Dissolved Solids (TDS) | | 736 | mg/L | 1 to 1,500 | >1,900 |
| Sodium Absorption Ratio (SAR) | | 1.9 | | <6.0 | >6.0 |
| SAR/EC Ratio (SEC) | | 1.7 | | <5.0 | >10.0 |
| pHc (pHc) | | 6.9 | | <8.4 may add Ca | >8.4 may remove Ca |

| Gypsum Requirements | 100% gypsum equivalent (lbs/ac-ft) |
|---|---|
| Eatons Gypsum Requirement (EGR) | 0 |
| Residual Sodium Carbonate (RSC) | 0 |

Sam Modesitt - Chemist

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# Ag Suitability Water Analysis

**LOUIE OTT & MORGAN WILLIS**
**GALLO FARMING, LP**
**P.O. BOX 775**
**ATWATER, CA 95301**

**Order Number:** 41113
**Lab Number:** W41113-13
**Submitted Date:** 12/6/2022
**Report Date:** 12/9/2022
**Submitted By:** SERGIO GARCIA
**COC Number:** WELL WATER

## Grower: JOSEPH GALLO FARMS

## Description: WELL 113-5

mg/L = milligrams/liter = part per million = ppm
meq/L = milliequivalents/liter
dS/m = deciSiemen/meter = mmhos/cm
lbs/ac-ft = pounds/acre-foot

| Analytes | | | | Normal Values | Problem Values |
|---|---|---|---|---|---|
| pH (pH) | | 7.9 | pH units | 6.5 to 8.0 | <6.5 or >8.0 |
| Electrical Conductivity (EC) | | 0.52 | dS/m | 0.5 to 3.0 | <0.5 or >3.0 |
| Boron (B) | | 0.04 | mg/L | <0.5 | >0.5 |
| **Cations** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Calcium (Ca) | 42.4 | 2.12 | 114 | 30 to 400 | <30 |
| Magnesium (Mg) | 9.7 | 0.81 | 26.1 | 1 to 60 | [Mg]>[Ca] |
| Sodium (Na) | 56.1 | 2.44 | 152 | <70 | >70 |
| Potassium (K) | 3.7 | 0.09 | 10.0 | | |
| **Anions** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Bicarbonate ($HCO_3$) | 217 | 3.56 | 586 | <150 | >300 |
| Chloride (Cl) | 17.3 | 0.49 | 46.7 | <150 | >200 |
| Nitrate - Nitrogen ($NO_3$-N) | 10.5 | 0.75 | 28.3 | <10 | >10 |
| Sulfate - Sulfur ($SO_4$-S) | 15.5 | 0.48 | 41.8 | | |
| **Calculated Values** | | | | **Normal Values** | **Problem Values** |
| Total Dissolved Solids (TDS) | | 358 | mg/L | 1 to 1,500 | >1,900 |
| Sodium Absorption Ratio (SAR) | | 2.0 | | <6.0 | >6.0 |
| SAR/EC Ratio (SEC) | | 3.9 | | <5.0 | >10.0 |
| pHc (pHc) | | 7.6 | | <8.4 may add Ca | >8.4 may remove Ca |

| Gypsum Requirements | 100% gypsum equivalent (lbs/ac-ft) |
|---|---|
| Eatons Gypsum Requirement (EGR) | 214 |
| Residual Sodium Carbonate (RSC) | 149 |

Sam Modesitt - Chemist

contact@vtaglab.com

RED = Value of Concern

**A & L Western Laboratories, Inc.**

1311 Woodland Avenue, Modesto CA 95351 209-529-4080
10220 SW Nimbus Avenue Bldg. K-9, Portland OR 97223 503-968-9225

# Ag Suitability Water Analysis

**LOUIE OTT & MORGAN WILLIS**
**GALLO FARMING, LP**
**P.O. BOX 775**
**ATWATER, CA 95301**

| | |
|---|---|
| **Order Number:** | **41113** |
| **Lab Number:** | **W41113-14** |
| **Submitted Date:** | **12/6/2022** |
| **Report Date:** | **12/9/2022** |
| **Submitted By:** | **SERGIO GARCIA** |
| **COC Number:** | **WELL WATER** |

## Grower: JOSEPH GALLO FARMS

## Description: WELL 92

mg/L = milligrams/liter = part per million = ppm
meq/L = milliequivalents/liter
dS/m = deciSiemen/meter = mmhos/cm
lbs/ac-ft = pounds/acre-foot

| Analytes | | | | Normal Values | Problem Values |
|---|---|---|---|---|---|
| pH (pH) | | 7.4 | pH units | 6.5 to 8.0 | <6.5 or >8.0 |
| Electrical Conductivity (EC) | | 1.91 | dS/m | 0.5 to 3.0 | <0.5 or >3.0 |
| Boron (B) | | 0.13 | mg/L | <0.5 | >0.5 |
| **Cations** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Calcium (Ca) | 155 | 7.74 | 418 | 30 to 400 | <30 |
| Magnesium (Mg) | 46.8 | 3.90 | 126 | 1 to 60 | [Mg]>[Ca] |
| Sodium (Na) | 195 | 8.47 | 526 | <70 | >70 |
| Potassium (K) | 6.1 | 0.16 | 16.5 | | |
| **Anions** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Bicarbonate ($HCO_3$) | 647 | 10.6 | 1750 | <150 | >300 |
| Chloride (Cl) | 149 | 4.19 | 401 | <150 | >200 |
| Nitrate - Nitrogen ($NO_3$-N) | 41.6 | 2.97 | 112 | <10 | >10 |
| Sulfate - Sulfur ($SO_4$-S) | 84.2 | 2.63 | 227 | | |
| **Calculated Values** | | | | **Normal Values** | **Problem Values** |
| Total Dissolved Solids (TDS) | | 1310 | mg/L | 1 to 1,500 | >1,900 |
| Sodium Absorption Ratio (SAR) | | 3.5 | | <6.0 | >6.0 |
| SAR/EC Ratio (SEC) | | 1.8 | | <5.0 | >10.0 |
| pHc (pHc) | | 6.7 | | <8.4 may add Ca | >8.4 may remove Ca |

| Gypsum Requirements | 100% gypsum equivalent (lbs/ac-ft) |
|---|---|
| Eatons Gypsum Requirement (EGR) | 0 |
| Residual Sodium Carbonate (RSC) | 0 |

Sam Modesitt - Chemist

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# Ag Suitability Water Analysis

**LOUIE OTT & MORGAN WILLIS**
**GALLO FARMING, LP**
**P.O. BOX 775**
**ATWATER, CA 95301**

**Order Number:** 41113
**Lab Number:** W41113-15
**Submitted Date:** 12/6/2022
**Report Date:** 12/9/2022
**Submitted By:** SERGIO GARCIA
**COC Number:** WELL WATER

## Grower: JOSEPH GALLO FARMS

## Description: WELL 91-5

mg/L = milligrams/liter = part per million = ppm
meq/L = milliequivalents/liter
dS/m = deciSiemen/meter = mmhos/cm
lbs/ac-ft = pounds/acre-foot

| Analytes | | | | Normal Values | Problem Values |
|---|---|---|---|---|---|
| pH (pH) | | 7.3 | pH units | 6.5 to 8.0 | <6.5 or >8.0 |
| Electrical Conductivity (EC) | | 1.41 | dS/m | 0.5 to 3.0 | <0.5 or >3.0 |
| Boron (B) | | 0.07 | mg/L | <0.5 | >0.5 |
| **Cations** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Calcium (Ca) | 120 | 6.02 | 325 | 30 to 400 | <30 |
| Magnesium (Mg) | 35.0 | 2.92 | 94.6 | 1 to 60 | [Mg]>[Ca] |
| Sodium (Na) | 147 | 6.38 | 396 | <70 | >70 |
| Potassium (K) | 5.3 | 0.14 | 14.4 | | |
| **Anions** | **mg/L** | **meq/L** | **lbs/ac-ft** | **Normal Values mg/L** | **Problem Values mg/L** |
| Bicarbonate ($HCO_3$) | 476 | 7.81 | 1290 | <150 | >300 |
| Chloride (Cl) | 93.2 | 2.63 | 252 | <150 | >200 |
| Nitrate - Nitrogen ($NO_3$-N) | 34.7 | 2.48 | 93.7 | <10 | >10 |
| Sulfate - Sulfur ($SO_4$-S) | 76.8 | 2.40 | 207 | | |
| **Calculated Values** | | | | **Normal Values** | **Problem Values** |
| Total Dissolved Solids (TDS) | | 970 | mg/L | 1 to 1,500 | >1,900 |
| Sodium Absorption Ratio (SAR) | | 3.0 | | <6.0 | >6.0 |
| SAR/EC Ratio (SEC) | | 2.1 | | <5.0 | >10.0 |
| pHc (pHc) | | 6.9 | | <8.4 may add Ca | >8.4 may remove Ca |

| Gypsum Requirements | 100% gypsum equivalent (lbs/ac-ft) |
|---|---|
| Eatons Gypsum Requirement (EGR) | 0 |
| Residual Sodium Carbonate (RSC) | 0 |

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